MATH READING NOTES: *Asymptotics: Geometric Optics, Stationary phase, and Homogenization*

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Based on the book *Princeton Lectures in Analysis: Fourier Analysis*
by Elias M. Stein and the book
Partial Differential Equations by Lawrence C. Evans

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Disclaimer: *These notes are informal, and have not been peer-reviewed, which may contain errors. Feedback is welcome (contact the author via email).* ♟️🐰

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1 Geometric Optics

1.1 Introduction to High-Frequency wave propagation

In the high-frequency regime ($\varepsilon \ll 1$), wave propagation can be approximated using *geometric optics*, where wavefronts and rays replace full wave solutions. The central tool is the *geometric optics ansatz*:

$$u^\varepsilon(x,t) = e^{ip^\varepsilon(x,t)/\varepsilon} a^\varepsilon(x,t). \tag{1.1}$$

where $p^\varepsilon(x,t)$: Real-valued phase function (rapidly varying due to $\varepsilon \ll 1$); $a^\varepsilon(x,t)$: Real-valued amplitude (slowly varying); $\varepsilon > 0$: Small parameter (wavelength scale).

Substituting the ansatz into the wave equation and taking $\varepsilon \to 0$ yields the *Hamilton-Jacobi equation* for the phase:

$$p_t \pm |\nabla p| = 0. \tag{1.2}$$

The characteristics of this PDE correspond to *light rays*, which describe the paths along which wave energy propagates. This links the asymptotic analysis to ray-tracing methods.

1.2 Physical Interpretation

1.2.1 Wavefronts and Rays

The geometric optics ansatz $u^\varepsilon = e^{ip^\varepsilon/\varepsilon} a^\varepsilon$ encodes two key physical quantities:

i. **Wavefronts**: Surfaces of constant phase ($p(x,t) = \text{const}$) that propagate at the *local phase velocity*. In the high-frequency limit ($\varepsilon \to 0$), these dominate the wave's spatial structure.

ii. **Rays**: Trajectories orthogonal to wavefronts (in isotropic media), representing the *direction of energy propagation*. These emerge as the characteristics of the Hamilton-Jacobi equation.

This phase-amplitude decoupling implies that wave energy localizes along rays (via amplitude a^ε), while wavefront geometry evolves via the phase p^ε (governed by (1.1)). Figure 1 illustrates this as below:

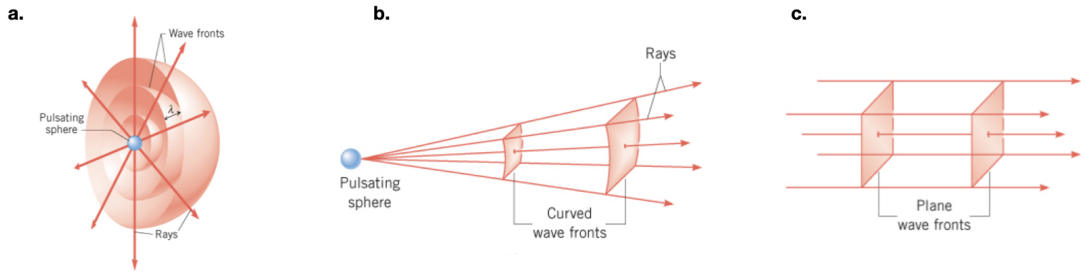


Figure 1: Wavefronts and rays in geometric optics. (a) Spherical wavefronts (spaced by wavelength λ) and diverging rays from a pulsating source. (b) Intermediate region: Decreasing curvature of expanding wavefronts. (c) Far-field limit: Nearly plane wavefronts with parallel rays [3].

1.2.2 *Huygens' Principle*

Huygens' principle states that each point on a wavefront at time t acts as a source of secondary spherical wavelets. The wavefront at $t + \Delta t$ is the tangential envelope of these wavelets. In the high-frequency limit ($\lambda \to 0$ or $k \to \infty$, where k is the wavenumber), this construction recovers geometric ray propagation.

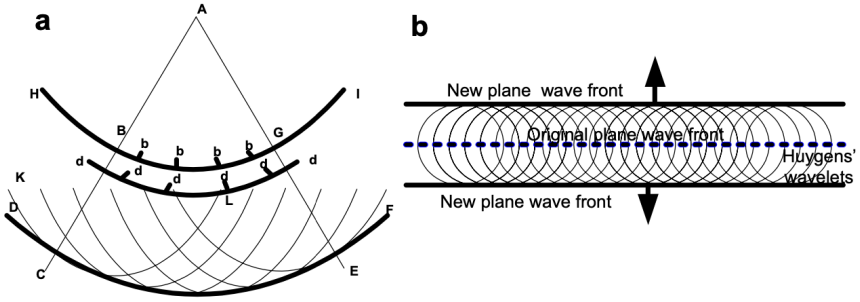


Figure 2: Huygens' Principle as a geometrical construction. (a) Spherical wavelets from a point source. (b) Plane wavefront construction [2].

While foundational, Huygens's Principle has key gaps:

- **Successes**: Predicts reflection/refraction (via wavefront continuity) and qualitatively explains diffraction.
- **Failures**: Lacks mechanisms for energy transport (resolved by Maxwell's equations) and quantum effects (e.g., photoelectric effect). Cannot describe polarization, absorption, or emission—phenomena later explained by electromagnetic theory and quantum optics [4].

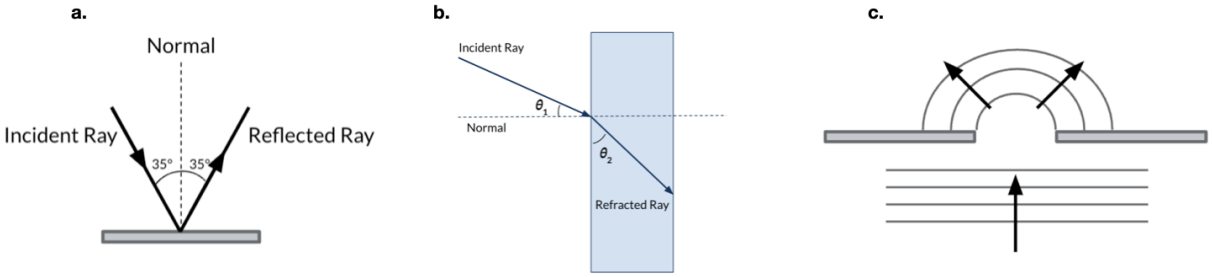


Figure 3: Three optical phenomena: (a) Reflection with $\theta_i = \theta_r = 35°$, (b) Refraction at an interface, and (c) Diffraction through a slit [1].

Summary: Geometric optics bridges wave and ray theories, offering a tractable high-frequency approximation. The ansatz connects asymptotic analysis to physical intuition, while Huygens' Principle provides historical context for modern ray-tracing methods.

1.3 Derivation of the Eikonal Equation

1. Step 1: Problem Setup:

To see how the physical interpretation relates to the mathematical theory, we consider the wave equation:

$$u_{tt} - \Delta u = 0 \quad \text{in } \mathbb{R}^n \times (0, \infty). \tag{1.3}$$

and seek high-frequency solutions using the *ansatz*: $u^\varepsilon(x,t) = e^{ip^\varepsilon(x,t)/\varepsilon} a^\varepsilon(x,t)$, where $p^\varepsilon(x,t)$: Rapidly varying phase ($\mathcal{O}(1)$ as $\varepsilon \to 0$), and $a^\varepsilon(x,t)$: Slowly varying amplitude.

2. Step 2: Compute Derivatives of the Ansatz:

Substitute u^ε into the wave equation by computing its time and spatial derivatives.

i. Time Derivatives: First time derivative (u_t^ε):

$$u_t^\varepsilon = \frac{\partial}{\partial t}\left(e^{ip^\varepsilon/\varepsilon} a^\varepsilon\right) = \frac{\partial}{\partial t}\left(e^{ip^\varepsilon/\varepsilon}\right) a^\varepsilon + e^{ip^\varepsilon/\varepsilon}\frac{\partial a^\varepsilon}{\partial t} = e^{ip^\varepsilon/\varepsilon}\left(\frac{ip_t^\varepsilon}{\varepsilon} a^\varepsilon + a_t^\varepsilon\right). \tag{1.4}$$

Second time derivative (u_{tt}^ε):

$$\begin{aligned} u_{tt}^\varepsilon &= \frac{\partial}{\partial t}\left[e^{ip^\varepsilon/\varepsilon}\left(\frac{ip_t^\varepsilon}{\varepsilon} a^\varepsilon + a_t^\varepsilon\right)\right] = \frac{\partial}{\partial t}\left(e^{ip^\varepsilon/\varepsilon}\right)\left(\frac{ip_t^\varepsilon}{\varepsilon} a^\varepsilon + a_t^\varepsilon\right) + e^{ip^\varepsilon/\varepsilon}\frac{\partial}{\partial t}\left(\frac{ip_t^\varepsilon}{\varepsilon} a^\varepsilon + a_t^\varepsilon\right) \\ &= e^{ip^\varepsilon/\varepsilon}\left[-\frac{(p_t^\varepsilon)^2}{\varepsilon^2} a^\varepsilon + \frac{ip_{tt}^\varepsilon}{\varepsilon} a^\varepsilon + \frac{2ip_t^\varepsilon a_t^\varepsilon}{\varepsilon} + a_{tt}^\varepsilon\right]. \end{aligned} \tag{1.5}$$

ii. Spatial Derivatives: Gradient (∇u^ε):

$$\nabla u^\varepsilon = \nabla\left(e^{ip^\varepsilon/\varepsilon} a^\varepsilon\right) = e^{ip^\varepsilon/\varepsilon}\left(\frac{i\nabla p^\varepsilon}{\varepsilon} a^\varepsilon + \nabla a^\varepsilon\right). \tag{1.6}$$

Laplacian (Δu^ε) [1]:

$$\begin{aligned} \Delta u^\varepsilon &= \nabla \cdot \left(e^{ip^\varepsilon/\varepsilon}\left(\frac{i\nabla p^\varepsilon}{\varepsilon} a^\varepsilon + \nabla a^\varepsilon\right)\right) \\ &= \nabla\left(e^{ip^\varepsilon/\varepsilon}\right) \cdot \left(\frac{i\nabla p^\varepsilon}{\varepsilon} a^\varepsilon + \nabla a^\varepsilon\right) + e^{ip^\varepsilon/\varepsilon}\nabla \cdot \left(\frac{i\nabla p^\varepsilon}{\varepsilon} a^\varepsilon + \nabla a^\varepsilon\right) \\ &= e^{ip^\varepsilon/\varepsilon}\left[-\frac{|\nabla p^\varepsilon|^2}{\varepsilon^2} a^\varepsilon + \frac{i\Delta p^\varepsilon}{\varepsilon} a^\varepsilon + \frac{2i\nabla p^\varepsilon \cdot \nabla a^\varepsilon}{\varepsilon} + \Delta a^\varepsilon\right]. \end{aligned} \tag{1.7}$$

Note: $e^{ip^\varepsilon/\varepsilon}$ appears in all terms. Derivatives of p^ε introduces powers of $1/\varepsilon$.

[1] The product rule for divergence: $\nabla \cdot (f\mathbf{F}) = \nabla f \cdot \mathbf{F} + f(\nabla \cdot \mathbf{F})$.

3. Step 3: Substitute into the Wave Equation:

Plug the derivatives into $u_{tt}^\varepsilon - \Delta u^\varepsilon = 0$ and divide by $e^{ip^\varepsilon/\varepsilon}$:

$$\left[-\frac{(p_t^\varepsilon)^2}{\varepsilon^2}a^\varepsilon + \frac{ip_{tt}^\varepsilon}{\varepsilon}a^\varepsilon + \frac{2ip_t^\varepsilon a_t^\varepsilon}{\varepsilon} + a_{tt}^\varepsilon\right] - \left[-\frac{|\nabla p^\varepsilon|^2}{\varepsilon^2}a^\varepsilon + \frac{i\Delta p^\varepsilon}{\varepsilon}a^\varepsilon + \frac{2i\nabla p^\varepsilon \cdot \nabla a^\varepsilon}{\varepsilon} + \Delta a^\varepsilon\right] = 0. \tag{1.8}$$

Group terms by powers of ε:

$$\underbrace{\left[-\frac{(p_t^\varepsilon)^2}{\varepsilon^2} + \frac{|\nabla p^\varepsilon|^2}{\varepsilon^2}\right]a^\varepsilon}_{\mathcal{O}(\varepsilon^{-2})} + \underbrace{\frac{i}{\varepsilon}\left(p_{tt}^\varepsilon - \Delta p^\varepsilon\right)a^\varepsilon + \frac{2i}{\varepsilon}\left(p_t^\varepsilon a_t^\varepsilon - \nabla p^\varepsilon \cdot \nabla a^\varepsilon\right)}_{\mathcal{O}(\varepsilon^{-1})} + \underbrace{\Delta a^\varepsilon}_{\mathcal{O}(1)} = 0. \tag{1.9}$$

- **Hierarchy of terms:**
  - $\mathcal{O}(\varepsilon^{-2})$: Phase terms (determine the eikonal equation).
  - $\mathcal{O}(\varepsilon^{-1})$: Phase-amplitude coupling (determine the transport equation for a).
  - $\mathcal{O}(1)$: Amplitude derivatives (negligible for leading-order asymptotics).

The ansatz assumes the amplitude a^ε varies much slower than the phase $p^\varepsilon/\varepsilon$, so its derivatives ($\nabla a^\varepsilon$, Δa^ε) do not scale with ε.

4. Step 4: Dominant Balance Analysis:

For the equation to hold, the coefficients of each power of ε must separately vanish.

i. Leading order $\mathcal{O}(\varepsilon^{-2})$: Recall we have:

$$-\frac{(p_t^\varepsilon)^2}{\varepsilon^2}a^\varepsilon + \frac{|\nabla p^\varepsilon|^2}{\varepsilon^2}a^\varepsilon = 0. \tag{1.10}$$

Since $a^\varepsilon \not\equiv 0$ (nontrivial solution), we divide by $a^\varepsilon/\varepsilon^2$, and take the limit $\varepsilon \to 0$ (assuming $p^\varepsilon \to p$ and $a^\varepsilon \to a \neq 0$):

$$-(p_t^\varepsilon)^2 + |\nabla p^\varepsilon|^2 = 0 \implies (p_t)^2 - |\nabla p|^2 = 0 \implies p_t \pm |\nabla p| = 0. \tag{1.11}$$

which is the eikonal equation.

ii. Next order $\mathcal{O}(\varepsilon^{-1})$: The $\mathcal{O}(\varepsilon^{-1})$ terms in the wave equation yield the transport equation for a^ε:

$$\underbrace{\frac{i}{\varepsilon}\left(p_{tt}^\varepsilon - \Delta p^\varepsilon\right)a^\varepsilon}_{\text{Phase curvature}} + \underbrace{\frac{2i}{\varepsilon}\left(p_t^\varepsilon a_t^\varepsilon - \nabla p^\varepsilon \cdot \nabla a^\varepsilon\right)}_{\text{Amplitude-phase coupling}} = 0. \tag{1.12}$$

Taking $\varepsilon \to 0$ (with $p^\varepsilon \to p, a^\varepsilon \to a$):

$$(p_{tt} - \nabla p)a + 2(p_t a_t - \nabla p \cdot \nabla a) = 0. \tag{1.13}$$

Note: The transport equation tells us about how the amplitude a evolves along the rays of the eikonal equation, ensuring energy conservation. The solution $a(x,t)$ depends on wavefront curvature (Δp) and propagation dynamics.

5. Step 5: The Eikonal Equation for General Hyperbolic PDE:

Consider a second-order hyperbolic PDE of the form:

$$u_{tt} - \sum_{k,l=1}^{n} a^{kl}(x) u_{x_k x_l} = 0. \tag{1.14}$$

where $a^{kl}(x) = a^{lk}(x)$ (symmetric coefficients), and the matrix (a^{kl}) is positive-definite (ensuring wave-like solutions). Again, we seek high-frequency solutions using the same ansatz $u^\varepsilon(x,t)$, with $p^\varepsilon = \mathcal{O}(1)$ and a^ε slowly varying as $\varepsilon \to 0$.

i. Compute Derivatives: First, we compute the necessary derivatives:

Time derivatives:

$$u_t^\varepsilon = e^{ip^\varepsilon/\varepsilon}\left(\frac{ip_t^\varepsilon}{\varepsilon}a^\varepsilon + a_t^\varepsilon\right). \tag{1.15}$$

$$u_{tt}^\varepsilon = \frac{\partial}{\partial t}\left[e^{ip^\varepsilon/\varepsilon}\left(\frac{ip_t^\varepsilon}{\varepsilon}a^\varepsilon + a_t^\varepsilon\right)\right] = e^{ip^\varepsilon/\varepsilon}\left[-\frac{(p_t^\varepsilon)^2}{\varepsilon^2}a^\varepsilon + \frac{ip_{tt}^\varepsilon}{\varepsilon}a^\varepsilon + \frac{2ip_t^\varepsilon}{\varepsilon}a_t^\varepsilon + a_{tt}^\varepsilon\right]. \tag{1.16}$$

Spatial derivatives:

$$u_{x_k}^\varepsilon = e^{ip^\varepsilon/\varepsilon}\left(\frac{ip_{x_k}^\varepsilon}{\varepsilon}a^\varepsilon + a_{x_k}^\varepsilon\right). \tag{1.17}$$

$$u_{x_k x_l}^\varepsilon = \frac{\partial}{\partial x_l}\left[e^{ip^\varepsilon/\varepsilon}\left(\frac{ip_{x_k}^\varepsilon}{\varepsilon}a^\varepsilon + a_{x_k}^\varepsilon\right)\right] = e^{ip^\varepsilon/\varepsilon}\left[-\frac{p_{x_k}^\varepsilon p_{x_l}^\varepsilon}{\varepsilon^2}a^\varepsilon + \frac{ip_{x_k x_l}^\varepsilon}{\varepsilon}a^\varepsilon + \frac{2ip_{x_k}^\varepsilon}{\varepsilon}a_{x_l}^\varepsilon + a_{x_k x_l}^\varepsilon\right]. \tag{1.18}$$

ii. Substitute into the PDE: Substituting into the original equation and dividing by $e^{ip^\varepsilon/\varepsilon}$ yields:

$$\left[-\frac{(p_t^\varepsilon)^2}{\varepsilon^2} + \frac{ip_{tt}^\varepsilon}{\varepsilon} + \frac{2ip_t^\varepsilon}{\varepsilon}\frac{a_t^\varepsilon}{a^\varepsilon} + \frac{a_{tt}^\varepsilon}{a^\varepsilon}\right] - \sum_{k,l} a^{kl}(x)\left[-\frac{p_{x_k}^\varepsilon p_{x_l}^\varepsilon}{\varepsilon^2} + \frac{ip_{x_k x_l}^\varepsilon}{\varepsilon} + \frac{2ip_{x_k}^\varepsilon}{\varepsilon}\frac{a_{x_l}^\varepsilon}{a^\varepsilon} + \frac{a_{x_k x_l}^\varepsilon}{a^\varepsilon}\right] = 0. \tag{1.19}$$

iii. Dominant Balance Analysis: Grouping terms by order of ε:

1. $\mathcal{O}(\varepsilon^{-2})$ terms (leading order):

$$-\frac{(p_t^\varepsilon)^2}{\varepsilon^2} + \frac{1}{\varepsilon^2}\sum_{k,l} a^{kl}(x) p_{x_k}^\varepsilon p_{x_l}^\varepsilon = 0. \tag{1.20}$$

Taking $\varepsilon \to 0$ gives the eikonal equation:

$$(p_t)^2 = \sum_{k,l} a^{kl}(x) p_{x_k} p_{x_l} \quad \text{or} \quad p_t = \pm\sqrt{\sum_{k,l} a^{kl}(x) p_{x_k} p_{x_l}}. \tag{1.21}$$

2. $\mathcal{O}(\varepsilon^{-1})$ terms (transport equation):

$$\frac{i}{\varepsilon}\left(p^{\varepsilon}_{tt}-\sum_{k,l}a^{kl}(x)p^{\varepsilon}_{x_kx_l}\right)+\frac{2i}{\varepsilon}\left(p^{\varepsilon}_{t}\frac{a^{\varepsilon}_{t}}{a^{\varepsilon}}-\sum_{k,l}a^{kl}(x)p^{\varepsilon}_{x_k}\frac{a^{\varepsilon}_{x_l}}{a^{\varepsilon}}\right)=0. \tag{1.22}$$

Taking $\varepsilon \to 0$ gives the transport equation:

$$2p_ta_t-2\sum_{k,l}a^{kl}(x)p_{x_k}a_{x_l}+\left(p_{tt}-\sum_{k,l}a^{kl}(x)p_{x_kx_l}\right)a=0. \tag{1.23}$$

Note: The eikonal equation $(p_t)^2=\sum a^{kl}(x)p_{x_k}p_{x_l}$ governs wavefront propagation in anisotropic media, and the transport equation describes amplitude evolution via advection and curvature effects for medium inhomogeneity. The isotropic case ($a^{kl}=\delta^{kl}$) recovers standard wave dynamics.

Remark: The condition $a^{kl}(x)>0$ ensures real characteristic (wave) speeds, energy conservation along rays, and well-posedness.

6. Step 6: Comparison between the Standard Wave Equation and the General Hyperbolic PDE:

Aspect	**Standard Wave Equation**	**General Hyperbolic PDE**
PDE	$u_{tt}-\Delta u=0$	$u_{tt}-\sum a^{kl}(x)u_{x_kx_l}=0$
Eikonal Equation	$p_t\pm\lvert Dp\rvert=0$	$p_t\pm\sqrt{\sum a^{kl}(x)p_{x_k}p_{x_l}}=0$
Rays	Straight lines	Curved (depends on $a^{kl}(x)$)
Amplitude Equation	Transport along rays	Modified by $a^{kl}(x)$

2 The Stationary Phase Method

2.1 Overview

The stationary phase method analyzes oscillatory integrals of the form:

$$I_\varepsilon = \int_{\mathbb{R}^n} e^{i\phi(y)/\varepsilon} a(y)\, dy \quad (\varepsilon \to 0). \tag{2.1}$$

where $\phi(y)$ is a real-valued phase function and $a(y) \in C_c^\infty(\mathbb{R}^n)$ is the amplitude. The method reveals that dominant contributions come from critical points where $\nabla\phi(y) = 0$, with other regions contributing negligibly due to rapid oscillations.

2.2 Key Lemmas

The mathematical foundation rests on three fundamental lemmas:

[Lemma 1 (Asymptotics for linear terms)] Let $a \in C_c^\infty(\mathbb{R}^n)$ and $p \in \mathbb{R}^n$, $p \neq 0$. Then for any integer $m \geq 1$, there exists a constant $C_m > 0$ such that:

$$\left| \int_{\mathbb{R}^n} e^{\frac{i}{\varepsilon} p\cdot y} a(y) dy \right| \leq C_m \varepsilon^m \quad \text{as } \varepsilon \to 0. \tag{2.2}$$

Note: This is the explicit meaning of the $O(\varepsilon^m)$ notation used in the source text. Proof: Repeated integration by parts yields the rapid decay.

[Lemma 2 (Quadratic Phase)] For $\phi(y) = \frac{1}{2} y^T A y$ (A symmetric, $\det A \neq 0$):

$$\int_{\mathbb{R}^n} e^{\frac{i}{2\varepsilon} y^T A y} a(y) dy = \frac{(2\pi\varepsilon)^{n/2}}{|\det A|^{1/2}} e^{\frac{i\pi}{4}\mathrm{sgn}(A)} (a(0) + O(\varepsilon)). \tag{2.3}$$

Proof: Exact Gaussian integral via diagonalization.

[Lemma 3 (Coordinate Changes)] The phase function can be simplified via a local change of coordinates:

i. **Non-stationary point ($\nabla\phi(y_0) \neq 0$):** ϕ can be made *affine*.

ii. **Non-degenerate critical point ($\nabla\phi(y_0) = 0$, $\det D^2\phi(y_0) \neq 0$):** ϕ can be made *purely quadratic* (Morse Lemma).

Note: The technical construction ensures $\Phi(y_0) = 0$ and $D\Phi(y_0) = I$, allowing direct application of Lemmas 1 and 2. (The complete version of Lemma 3 and proofs can be found at the Sec (2.5))

Summary: The stationary phase method analyzes oscillatory integrals via three steps: (1) rapid decay from linear phases, (2) Gaussian asymptotics from quadratic phases, and (3) coordinate reduction of general phases to these cases. This isolates contributions near critical points, underpinning high-frequency asymptotics in waves, optics, and quantum mechanics. This logical procedure is summarized in the decision flowchart below.

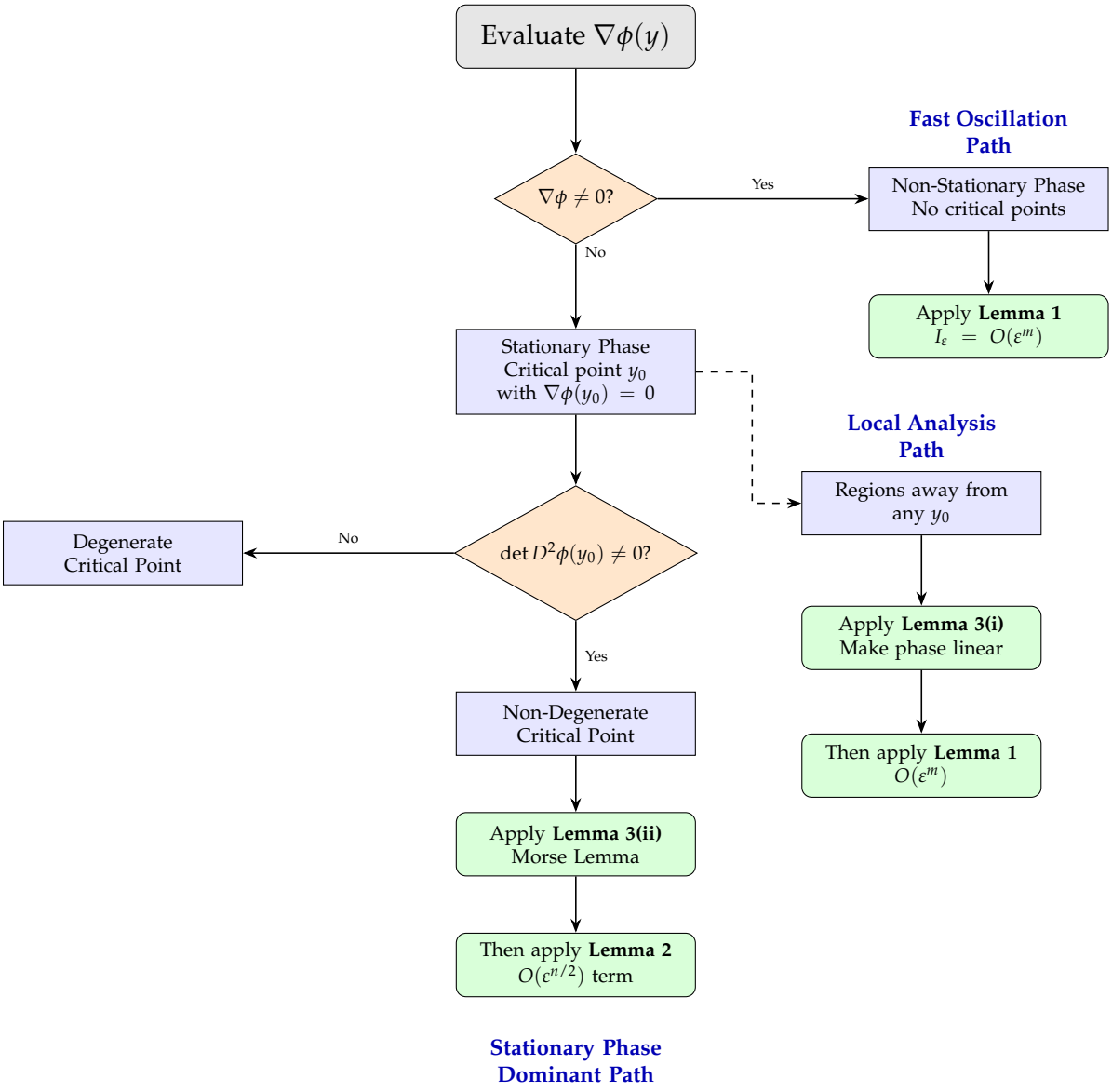


Figure 4: **Logical flow of the stationary phase method**. The full integral is solved using a partition of unity. Regions *away from* critical points are handled by Lemma 3(i) $\rightarrow$ Lemma 1. Regions *near* critical points contribute as shown in the dominant path.

Note: The case of *degenerate* critical points is not covered by the presented lemmas and requires more advanced techniques. It is included in the diagram for conceptual completeness but is beyond the scope of this analysis.

2.3 Applying Stationary Phase to the Wave Equation

We consider the high-frequency initial value problem for the wave equation:

$$\begin{cases} \partial_{tt} u^\varepsilon - \Delta u^\varepsilon = 0, & x \in \mathbb{R}^n, t > 0 \\ u^\varepsilon(x,0) = a(x)e^{ip(x)/\varepsilon} = g^\varepsilon(x), & a, p \in C_c^\infty(\mathbb{R}^n) \\ \partial_t u^\varepsilon(x,0) = 0. & \nabla p(x) \neq 0 \text{ on } \operatorname{supp}(a) \end{cases} \tag{2.4}$$

Here $a(x)$ is a smooth, compactly supported amplitude, $p(x)$ is a smooth phase function, and $\varepsilon \ll 1$ is the high-frequency parameter.

2.3.1 Fourier Transform and Solution Representation

For the system (2.4), we want to see how the initial position changes. We do a Fourier Transform:

$$u^\varepsilon(x,0) = a(x)e^{ip(x)/\varepsilon} = g^\varepsilon(x) \implies \widehat{g}^\varepsilon(\xi) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} a(x)e^{ip(x)/\varepsilon} e^{-i\xi x} dx. \tag{2.5}$$

This resolves the initial wave into its constituent plane waves $e^{i\xi\cdot x}$, with $\widehat{g}^\varepsilon(\xi)$ representing their complex amplitude. The general solution of the Cauchy problem for the wave equation [2]is:

$$\partial_{tt} u^\varepsilon - \nabla u^\varepsilon = 0 \implies u^\varepsilon(x,t) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \left(\widehat{g}^\varepsilon(\xi)\cos(t|\xi|) + \widehat{h}^\varepsilon(\xi)\frac{\sin(t|\xi|)}{|\xi|} \right) e^{i\xi x} d\xi. \tag{2.6}$$

where $\widehat{g}^\varepsilon(\xi)$ is the Fourier transform of $u^\varepsilon(x,0)$, and $\widehat{h}^\varepsilon(\xi)$ is the Fourier transform of $\partial_t u^\varepsilon(x,0)$. Moreover, since $\partial_t u^\varepsilon(x,0) = 0$, we have $\widehat{h}^\varepsilon(\xi) = 0$. This gives us the below expression:

$$\begin{aligned} u^\varepsilon(x,t) &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \widehat{g}^\varepsilon(\xi)\cos(t|\xi|)e^{i\xi\cdot x} d\xi. \\ &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \left(\frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} a(z)e^{ip(z)/\varepsilon} e^{-i\xi\cdot z} dz \right) \cos(t|\xi|)e^{i\xi\cdot x} d\xi. \\ &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)e^{ip(z)/\varepsilon} e^{i\xi\cdot(x-z)} \cos(t|\xi|) dz d\xi. \quad \text{(Fubini's Theorem)} \end{aligned} \tag{2.7}$$

Let $\cos(t|\xi|) = \frac{e^{it|\xi|} + e^{-it|\xi|}}{2}$, we decompose the solution into forward and backward propagating components:

$$\begin{aligned} u^\varepsilon(x,t) &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)e^{ip(z)/\varepsilon} e^{i\xi(x-z)} \left(\frac{e^{it|\xi|} + e^{-it|\xi|}}{2} \right) dz d\xi \\ &= \frac{1}{2}\left(\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)e^{ip(z)/\varepsilon} e^{i\xi(x-z)} e^{it|\xi|} dz d\xi \right) + \frac{1}{2}\left(\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)e^{ip(z)/\varepsilon} e^{i\xi(x-z)} e^{-it|\xi|} dz d\xi \right). \end{aligned} \tag{2.8}$$

[2]See Theorem 3.1 in [5].

This representation shows that $u^\varepsilon(x,t)$ depends on (1) time t, (2) the observation point x, and (3) the high-frequency parameter ε. Moreover, the two integrals represent the forward (I_+) and backward (I_-) propagating waves, respectively:

$$I_\pm^\varepsilon(x,t) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)e^{ip(z)/\varepsilon}e^{i\xi\cdot(x-z)}e^{\pm it|\xi|}dz\,d\xi. \tag{2.9}$$

To cast the integral in a form suitable for the stationary phase method, we rescale the wavevector. Let $\xi = \eta/\varepsilon$, so that $d\xi = d\eta/\varepsilon^n$. Substituting yields:

$$\begin{aligned} I_\pm^\varepsilon(x,t) &= \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)e^{ip(z)/\varepsilon}e^{i(\eta/\varepsilon)\cdot(x-z)}e^{\pm it|\eta/\varepsilon|}\frac{d\eta}{\varepsilon^n}dz \\ &= \frac{1}{(2\pi\varepsilon)^n}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)\exp\left(\frac{i}{\varepsilon}\left[p(z)+\eta\cdot(x-z)\pm t|\eta|\right]\right)dz\,d\eta. \end{aligned} \tag{2.10}$$

Relabeling the dummy variable η back to ξ gives the final form:

$$I_\pm^\varepsilon(x,t) = \frac{1}{(2\pi\varepsilon)^n}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} a(z)\exp\left(\frac{i}{\varepsilon}\left[p(z)+\xi\cdot(x-z)\pm t|\xi|\right]\right)dz\,d\xi. \tag{2.11}$$

Here, I_+ is the forward-propagating wave; I_- is the backward-propagating wave. The variable $z \in \mathbb{R}^n$ is a spatial variable (source point), while $\xi \in \mathbb{R}^n$ is a frequency variable (wavevector).

The form of (2.11) allows us to identify the key phase function that governs the high-frequency oscillations:

$$\phi_\pm(x,z,\xi,t) = \underbrace{p(z)}_{①} + \underbrace{\xi\cdot(x-z)}_{②} \pm \underbrace{t|\xi|}_{③}. \tag{2.12}$$

The terms have a clear physical interpretation:

① **Initial phase:** The value of the phase function $p(z)$ at the source point z.

② **Spatial propagation:** The phase shift for a wave with wavevector ξ propagating from z to x.

③ **Temporal evolution:** The phase shift $\pm\omega t$ due to time evolution, where $\omega = |\xi|$ is the frequency.

2.4 The Stationary Phase Method

Physical Intuition: For low frequency ($\varepsilon \sim 1$), wave propagation is resolved (individual oscillations visible). However, for high frequency ($\varepsilon \to 0$), only the envelope (wavefronts) is observable; oscillations are "averaged out." Moreover, High-frequency waves are governed by stationary points ($\nabla\phi = 0$), where oscillations constructively interfere; elsewhere, variations "cancel out."

Stationary Phase Analysis: We consider an oscillatory integral of the form:

$$I_\varepsilon = \int_{\mathbb{R}^n} a(y)e^{i\phi(y)/\varepsilon}dy. \tag{2.13}$$

where $a(y) \in C_c^\infty(\mathbb{R}^n)$ is the amplitude, $\phi(y)$ is a real-valued phase function, and $\varepsilon > 0$ is a small parameter. We want to understand the asymptotic behavior as $\varepsilon \to 0$.

As mentioned in **Fig 4**, we have the stationary and non-stationary cases. We start with the non-stationary phase first, where it can be branched out into the linear, nonlinear and general cases. Moreover, we give two examples respectively with the linear and nonlinear situations.

2.4.1 Case 1: Non-Stationary Phase.

[Case 1-i: Non-Stationary Phase ($\nabla\phi \neq 0$), the linear case] We first examine the fundamental case of a linear phase with a fixed, unit-frequency direction:

$$\phi(y) = p \cdot y, \quad \text{where } p \in S^{n-1}{}^a \text{ and } p \neq 0. \tag{2.14}$$

For this phase, direct integration by parts yields:

$$\left| \int_{\mathbb{R}^n} e^{\frac{i}{\varepsilon} p \cdot y} a(y) dy \right| \leq C_m \varepsilon^m \quad \text{as } \varepsilon \to 0. \tag{2.15}$$

Note: This is a special case of Lemma 1. The general result for any $p \in \mathbb{R}^n$, $p \neq 0$ follows by writing $p = |p|\tilde{p}$ with $\tilde{p} \in S^{n-1}$, which rescales $\varepsilon \mapsto \varepsilon/|p|$. The decay rate remains $O(\varepsilon^m)$, though the constant C_m depends on $|p|$.

[a] S^{n-1} is the $(n-1)$-dimensional unit sphere in $\mathbb{R}^n$, defined as $\{p \in \mathbb{R}^n \mid |p| = 1\}$.

1. *Step 1: Single Integration by Parts:*
   Without loss of generality, assume $p_1 \neq 0$. Note that $e^{ip\cdot y/\varepsilon} = \frac{\varepsilon}{ip_1}\partial_{y_1} e^{ip\cdot y/\varepsilon}$. Therefore:

$$I_\varepsilon = \int_{\mathbb{R}^n} a(y) e^{ip\cdot y/\varepsilon} dy = \frac{\varepsilon}{ip_1} \int_{\mathbb{R}^n} a(y) \partial_{y_1} e^{ip\cdot y/\varepsilon} dy. \tag{2.16}$$

   Applying integration by parts once yields:

$$\begin{aligned} I_\varepsilon &= \frac{\varepsilon}{ip_1} \left[a(y) e^{ip\cdot y/\varepsilon} \Big|_{y_1=-\infty}^{\infty} - \int_{\mathbb{R}^n} (\partial_{y_1} a(y)) e^{ip\cdot y/\varepsilon} dy \right] \\ &= -\frac{\varepsilon}{ip_1} \int_{\mathbb{R}^n} (\partial_{y_1} a(y)) e^{ip\cdot y/\varepsilon} dy. \end{aligned} \tag{2.17}$$

   The boundary term vanishes because a has compact support.

2. *Step 2: Iterate m Times:*
   Repeating this integration by parts process k times transfers all derivatives to the amplitude function:

$$I_\varepsilon = \left(-\frac{\varepsilon}{ip_1} \right)^m \int_{\mathbb{R}^n} (\partial_{y_1}^m a(y)) e^{ip\cdot y/\varepsilon} dy. \tag{2.18}$$

3. *Step 3: Bound the Integral:*
   Since $a \in C_c^\infty$, its derivatives are bounded and it has compact support. Therefore,

$$|I_\varepsilon| \leq \left(\frac{\varepsilon}{|p_1|}\right)^m \int_{\text{supp}(a)} |\partial_{y_1}^m a(y)| dy = \frac{\|\partial_{y_1}^m a\|_{L^1}}{|p_1|^m} \varepsilon^k = C_m \varepsilon^m. \tag{2.19}$$

   Hence, for any $m \geq 1$, $|I_\varepsilon| \leq C_m \varepsilon^m \to 0$ as $\varepsilon \to 0$.

Physical Interpretation: Rapid oscillations ($e^{ip\cdot y/\varepsilon}$) cause destructive interference, leading to the integral decaying faster than any power of ε.

Note: This result provides the detailed proof of **Lemma 1** (Asymptotics for linear terms) 2.2. Moreover, it also shows the **Fast Oscillation Path** from the **Fig 4: Logical flow of the stationary phase method** ☺.

[Case 1-ii: Non-Stationary Phase ($\nabla\phi \neq 0$), the non-linear case] We analyze the phase:

$$\phi(y) = p \cdot y + |y|^2 h(y), \quad p \neq 0. \tag{2.20}$$

We will show that under a key assumption, its gradient is bounded below:

$$|\nabla\phi(y)| \geq |p|/2, \quad \text{for } y \in \text{supp}(a). \tag{2.21}$$

Key Assumption: $p \cdot (2yh(y) + |y|^2 \nabla h(y)) \geq 0$ for all $y \in \text{supp}(a)$. (This means the non-linear term never opposes the direction of p).

This specific case can be handled directly. For the general case, we will later use a partition of unity to separate the integral into **non-critical regions** (handled by Lemma 3(i)) and **critical regions**. Here, for the non-stationary phase situation, we focus on the non-critical regions.

1. *Step 1: Compute the Gradient:*
   Recall the phase function $\phi(y)$ consists of a linear term and higher-order corrections:

$$\phi(y) = \underbrace{p \cdot y}_{\text{linear}} + \underbrace{|y|^2 h(y)}_{\text{nonlinear}}, \quad |y|^2 = y_1^2 + y_2^2 + y_3^2 + \cdots y_n^2 = \sum_i^n y_i^2. \tag{2.22}$$

   To see how it changes, we know its gradient is:

$$\begin{aligned} \nabla\phi(y) &= \nabla(p \cdot y) + \nabla(|y|^2 h(y)) = p + h(y)\nabla(|y|^2) + |y|^2 \nabla h(y) \\ &= p + 2yh(y) + |y|^2 \nabla h(y). \end{aligned} \tag{2.23}$$

   The critical points occur where $\nabla\phi(y) = 0$. Away from critical points, we want to establish a lower bound on $|\nabla\phi(y)|$.

2. *Step 2: Prove the Lower Bound (Under Key Assumption):*
   Compute the magnitude squared:

$$|\nabla\phi(y)|^2 = |p|^2 + |2yh(y) + |y|^2\nabla h(y)|^2 + 2p \cdot (2yh(y) + |y|^2\nabla h(y)). \tag{2.24}$$

   The key assumption ensures the third term is non-negative. Therefore:

$$|\nabla\phi(y)|^2 \geq |p|^2 \implies |\nabla\phi(y)| \geq |p| \geq |p|/2. \tag{2.25}$$

   This proves the bound.

3. *Step 3: Perform Integration by Parts via a Differential Operator:*
   The direct integration by parts used in **Case 1-i** relied on a specific coordinate direction (y_1 aligned with p). For a general nonlinear phase, we need a method that works in any direction. This is achieved by constructing a differential operator L tailored to the phase function ϕ [3]

   3.1 **Define the Operator:** We define the first-order differential operator

$$L := \frac{1}{i|\nabla\phi(y)|^2}\nabla\phi(y) \cdot \nabla = \frac{\nabla\phi}{i|\nabla\phi|^2} \cdot \nabla. \tag{2.26}$$

   The vector $\frac{\nabla\phi}{|\nabla\phi|^2}$ is well-defined because $|\nabla\phi(y)| \geq |p|/2 > 0$.

   3.2 **Key Property of L:** The operator L is designed to have a crucial property when applied to the oscillatory exponential:

$$\begin{aligned} L\left(e^{i\phi(y)/\varepsilon}\right) &= \frac{\nabla\phi}{i|\nabla\phi|^2} \cdot \nabla\left(e^{i\phi(y)/\varepsilon}\right) = \frac{\nabla\phi}{i|\nabla\phi|^2} \cdot \left(\frac{i}{\varepsilon}\nabla\phi\, e^{i\phi/\varepsilon}\right) \\ &= \frac{1}{\varepsilon}\frac{|\nabla\phi|^2}{|\nabla\phi|^2}e^{i\phi/\varepsilon} = \frac{1}{\varepsilon}e^{i\phi/\varepsilon}. \end{aligned} \tag{2.27}$$

   Thus, we can write $e^{i\phi/\varepsilon} = \varepsilon\, L(e^{i\phi/\varepsilon})$.

   3.3 **Integration by Parts (Formal Adjoint):** We now use this identity in the integral and integrate by parts. The formal adjoint of L, denoted L^*, is defined such that for any test functions f, g:

$$\int (Lf)g\,dy = \int f(L^*g)dy. \tag{2.28}$$

   holds, provided boundary terms vanish (which they do, as a has compact support). The adjoint is found by moving the derivatives onto the other function:

$$L^* = -\nabla \cdot \left(\frac{\nabla\phi}{i|\nabla\phi|^2}\right). \tag{2.29}$$

[3] Details and proofs are reviewed in reading notes [6].

Note the negative sign and the divergence replacing the gradient.

3.4 **Apply the Operator m Times:** We can now iteratively apply this process:

$$I_\varepsilon = \int a(y)e^{i\phi/\varepsilon}dy = \int a(y)\left(\varepsilon\, L(e^{i\phi/\varepsilon})\right)dy. = \varepsilon\int\left(L(e^{i\phi/\varepsilon})\right)a(y)dy$$
$$= \varepsilon\int e^{i\phi/\varepsilon}\left(L^*a(y)\right)dy \quad \text{since} \int (Lf)\,g\,dy = \int f\,(L^*g)\,dy. \tag{2.30}$$

We have effectively transferred the action from the exponential to the amplitude. Repeating this process m times yields:

$$I_\varepsilon = \varepsilon^m \int e^{i\phi/\varepsilon}\left((L^*)^m a(y)\right)dy. \tag{2.31}$$

3.5 **Final Estimate:** The function $(L^*)^m a(y)$ is a new amplitude. Since $a \in C_c^\infty(\mathbb{R}^n)$ and $|\nabla\phi|$ is bounded away from zero, $(L^*)^m a$ is also smooth and compactly supported. Define $C_m := \int |(L^*)^m a(y)|dy < \infty$. Therefore:

$$|I_\varepsilon| = \left|\varepsilon^m \int e^{i\phi/\varepsilon}(L^*)^m a(y)dy\right| \le \varepsilon^m \int \left|e^{i\phi/\varepsilon}\right| |(L^*)^m a(y)|\,dy$$
$$= \varepsilon^m \int |(L^*)^m a(y)|\,dy \quad (\text{since } |e^{i\theta}| = 1)$$
$$= C_m\varepsilon^m. \tag{2.32}$$

This proves that $|I_\varepsilon| \le C_m\varepsilon^m$ for any integer $m \ge 1$, meaning the integral decays faster than any power of ε. We recover **Lemma 1** (Asymptotics for linear terms) 2.2 result, which leads the **Fast Oscillation Path** from the **Fig 4: Logical flow of the stationary phase method** ☺.

The General Case via Lemma 3(i) and a Partition of Unity:
The IBP argument from Step 3 fails where $|\nabla\phi|$ is small. To overcome this, we use a partition of unity to localize the analysis. On regions without critical points, **Lemma 3(i)** (2.5) provides a local coordinate system Φ_α that linearizes the phase: $\phi(\Phi_\alpha(x)) = \phi(y_\alpha) + \nabla\phi(y_\alpha)\cdot x$.

4.1 **Localize with a Partition of Unity:** We cover $\text{supp}(a)$ with open sets $\{U_k\}_{k=0}^N$ such that:

i. For $k = 1, ..., N$, each U_k contains exactly one critical point y_k ($\nabla\phi(y_k) = 0$).

ii. U_0 contains no critical points ($|\nabla\phi| > 0$ on U_0).

Let $\{\psi_k\}_{k=0}^N$ be a smooth partition of unity subordinate to this cover ($0 \le \psi_k \le 1$, $\sum\psi_k = 1$ on $\text{supp}(a)$, $\text{supp}(\psi_k) \subset U_k$). This decomposes the integral:

$$I_\varepsilon = \int a(y)e^{i\phi/\varepsilon}dy = \sum_{k=0}^N \int \psi_k(y)a(y)e^{i\phi/\varepsilon}dy = \sum_{k=0}^N I_\varepsilon^{(k)}. \tag{2.33}$$

Our goal is now to analyze each $I_\varepsilon^{(k)}$ separately.

4.2 **Apply Lemma 3(i) on Non-Critical Regions:** We now analyze $I_\varepsilon^{(0)} = \int \psi_0(y)a(y)e^{i\phi/\varepsilon}dy$ where U_0 has no critical points. The challenge is that $|\nabla\phi|^{-1}$ may be unbounded on U_0. Our strategy is a second, finer localization:

4.2.1 **Create a Fine Cover:** Cover U_0 with sets $\{V_\alpha\}$ small enough for **Lemma 3(i)** to apply on each:

$$\phi(\Phi_\alpha(x)) = \phi(y_\alpha) + \nabla\phi(y_\alpha)\cdot x. \tag{2.34}$$

4.2.2 **Analyze a Single Patch:** On one patch V_α, the change of variables yields [4]:

$$\begin{aligned}\int_{V_\alpha} \psi_0(y)a(y)e^{i\phi(y)/\varepsilon}dy &= \int_{\Phi_\alpha^{-1}(V_\alpha)} \psi_0(\Phi_\alpha(x))a(\Phi_\alpha(x))e^{i\phi(\Phi_\alpha(x))/\varepsilon}|\det D\Phi_\alpha(x)|dx \\ &= \int_{\Phi_\alpha^{-1}(V_\alpha)} \psi_0(\Phi_\alpha(x))a(\Phi_\alpha(x))e^{i[\phi(y_\alpha)+\nabla\phi(y_\alpha)\cdot x]/\varepsilon}|\det D\Phi_\alpha(x)|dx \\ &= e^{i\phi(y_\alpha)/\varepsilon}\int_{\Phi_\alpha^{-1}(V_\alpha)} \underbrace{\psi_0(\Phi_\alpha(x))a(\Phi_\alpha(x))|\det D\Phi_\alpha(x)|}_{\text{New amplitude on patch}} e^{i[\nabla\phi(y_\alpha)\cdot x]/\varepsilon}dx.\end{aligned} \tag{2.35}$$

4.2.3 **Recombine with a Second Partition of Unity:** To reconstruct $I_\varepsilon^{(0)}$ rigorously, use the P.o.U. $\{\tilde\psi_\alpha\}$ subordinate to $\{V_\alpha\}$:

$$\begin{aligned}I_\varepsilon^{(0)} &= \int_{U_0} \psi_0(y)a(y)e^{i\phi(y)/\varepsilon}dy = \int_{U_0}\left(\sum_\alpha \tilde\psi_\alpha(y)\right)\psi_0(y)a(y)e^{i\phi(y)/\varepsilon}dy \\ &= \sum_\alpha \int_{U_0} \tilde\psi_\alpha(y)\psi_0(y)a(y)e^{i\phi(y)/\varepsilon}dy \quad \text{(Interchange sum and integral)} \\ &= \sum_\alpha \int_{V_\alpha} \tilde\psi_\alpha(y)\psi_0(y)a(y)e^{i\phi(y)/\varepsilon}dy. \quad \text{(Since } \operatorname{supp}(\tilde\psi_\alpha)\subset V_\alpha\text{)}\end{aligned} \tag{2.36}$$

4.2.4 **Apply Change of Variables to Each Term:** Now apply $y = \Phi_\alpha(x)$ to each term in the sum:

$$\begin{aligned}I_\varepsilon^{(0)} &= \sum_\alpha \int \tilde\psi_\alpha(\Phi_\alpha(x))\psi_0(\Phi_\alpha(x))a(\Phi_\alpha(x))e^{i[\phi(y_\alpha)+\nabla\phi(y_\alpha)\cdot x]/\varepsilon}|\det D\Phi_\alpha(x)|dx \\ &= \sum_\alpha \int \tilde\psi_\alpha(x)\underbrace{a(\Phi_\alpha(x))|\det D\Phi_\alpha(x)|}_{\tilde a_\alpha(x)} e^{i(\phi(y_\alpha)+\nabla\phi(y_\alpha)\cdot x)/\varepsilon}dx.\end{aligned} \tag{2.37}$$

The function $\tilde\psi_\alpha(x)$ ensures the integration is restricted to the region corresponding to V_α.

4.3 **Establish the Final Bound:** We now show that $I_\varepsilon^{(0)} = O(\varepsilon^m)$ by bounding the final expression from Step 4.2.4.

4.3.1 **Recognize the Linear Phase Form:** Crucially, on the support of each $\tilde\psi_\alpha$, the phase

[4] Multidimensional change of variables: $dy = |\det D\Phi(x)|dx$. The determinant accounts for local volume distortion under the coordinate transformation $y = \Phi(x)$.

is *linear* with $\nabla\phi(y_\alpha) \neq 0$. Therefore, each integral in the sum

$$J_\alpha := \int \tilde{\psi}_\alpha(x)\tilde{a}_\alpha(x)e^{i(\phi(y_\alpha)+\nabla\phi(y_\alpha)\cdot x)/\varepsilon}dx. \tag{2.38}$$

has exactly the form studied in **Lemma 1** (2.2) (with amplitude $\tilde{\psi}_\alpha(x)\tilde{a}_\alpha(x)$ and linear phase $\nabla\phi(y_\alpha)\cdot x$).

4.3.2 **Apply the Linear Phase Result:** Applying the result from **Lemma 1** (2.2), we conclude that for any integer $m \geq 1$, there exists a constant $C_{m,\alpha} > 0$ such that:

$$|J_\alpha| = \left| \int \tilde{\psi}_\alpha(x)\tilde{a}_\alpha(x)e^{i(\phi(y_\alpha)+\nabla\phi(y_\alpha)\cdot x)/\varepsilon}dx \right| \leq C_{m,\alpha}\varepsilon^m. \tag{2.39}$$

The constant $C_{m,\alpha}$ depends on m, the patch V_α, and the properties of the amplitude function $\tilde{\psi}_\alpha(x)\tilde{a}_\alpha(x)$.

4.3.3 **Combine the Finite Sum:** Since the number of patches α is finite, we may define $C_m := \sum_\alpha C_{m,\alpha} < \infty$.

4.3.4 **Apply the Triangle Inequality:** Taking the absolute value of the full sum and applying the triangle inequality yields the final estimate:

$$|I_\varepsilon^{(0)}| = \left| \sum_\alpha J_\alpha \right| \leq \sum_\alpha |J_\alpha| \leq \sum_\alpha C_{m,\alpha}\varepsilon^m = C_m\varepsilon^m. \tag{2.40}$$

This proves that $I_\varepsilon^{(0)} = O(\varepsilon^m)$ for any $m \geq 1$. This also shows the **Local Analysis Path** from the **Fig 4: Logical flow of the stationary phase method** ☺.

While the preceding cases (linear and specific nonlinear phases) demonstrate the core idea of destructive interference, the true power of the method lies in its general applicability. We now present the general proof for the decay of oscillatory integrals with non-stationary phase.

[Case 1-iii: Non-Stationary Phase ($\nabla\phi \neq 0$), the general case] The goal is to show that when $\nabla\phi \neq 0$ on $\text{supp}(a)$, the oscillatory integral:

$$I_\varepsilon = \int_{\mathbb{R}^n} a(y)e^{i/\varepsilon\phi(y)}dy, \quad a \in C_c^\infty(\mathbb{R}^n). \tag{2.41}$$

decays faster than any power of ε as $\varepsilon \to 0^+$.

1. *Step 1: Rewrite the Oscillatory Factor:* We assume that the amplitude $a \in C_c^\infty(\mathbb{R}^n)$ (smooth with compact support), the phase $\phi \in C^\infty(\mathbb{R}^n)$ with $|\nabla\phi(y)| \geq C > 0$ for $y \in \text{supp}(a)$. Moreover, the small parameter $\varepsilon \to 0^+$ for the high frequency limit.

   We want to investigate on the oscillatory part. We express $e^{i\phi(y)/\varepsilon}$ in terms of its own derivative, and define a directional vector field $v(y)$:

$$\nabla(e^{i\phi(y)/\varepsilon}) = \frac{i}{\varepsilon}\nabla\phi(y)e^{i\phi(y)/\varepsilon}, \quad v(y) = \frac{\nabla\phi(y)}{|\nabla\phi(y)|^2}. \tag{2.42}$$

Now, put them together:

$$v(y)\cdot\nabla(e^{i\phi(y)/\varepsilon}) = \frac{\nabla\phi(y)}{|\nabla\phi(y)|^2}\cdot\frac{i}{\varepsilon}\nabla\phi(y)e^{i\phi(y)/\varepsilon} = \frac{i}{\varepsilon}e^{i\phi(y)/\varepsilon}. \tag{2.43}$$

This gives us:

$$e^{i\phi(y)/\varepsilon} = \frac{\varepsilon}{i}v(y)\cdot\nabla(e^{i\phi(y)/\varepsilon}). \tag{2.44}$$

2 *Step 2: Integration by Parts:* We substitute it back to the integral:

$$I_\varepsilon = \int_{\mathbb{R}^n} a(y)\left(\frac{\varepsilon}{i}v(y)\cdot\nabla(e^{i\phi(y)/\varepsilon})\right)dy = \frac{\varepsilon}{i}\int_{\mathbb{R}^n} a(y)v(y)\cdot\nabla e^{i\phi(y)/\varepsilon}dy. \tag{2.45}$$

Then, we do integration by parts, with assuming no boundary terms:

$$\int_{\mathbb{R}^n} F\cdot\nabla g dy = -\int_{\mathbb{R}^n}(\text{divF})g dy. \tag{2.46}$$

Let $F(y) = a(y)v(y)$ and $g(y) = e^{i\phi(y)/\varepsilon}$, we have:

$$I_\varepsilon = \frac{\varepsilon}{i}\left(-\int_{\mathbb{R}^n}\text{div}(a(y)v(y))e^{i\phi(y)/\varepsilon}\right)dy = i\varepsilon\int_{\mathbb{R}^n}\text{div}\left(a(y)\frac{\nabla\phi(y)}{|\nabla\phi(y)|^2}\right)e^{i\phi(y)/\varepsilon}dy. \tag{2.47}$$

3. *Step 3: Iterative Scheme:* Recall we have (2.47), which can be viewed as:

$$I_\varepsilon = i\varepsilon\int\underbrace{\text{div}(a(y)v(y))}_{a_1(y)}e^{i\phi(y)/\varepsilon}dy, \text{ where } v(y) = \frac{\nabla\phi(y)}{|\nabla\phi(y)|^2}. \tag{2.48}$$

Now, we apply the same trick to the new integral:

$$\int a_1(y)e^{i\phi(y)/\varepsilon}dy = i\varepsilon\int\text{div}(a_1(y)v(y))e^{i\phi(y)/\varepsilon}dy. \implies I_\varepsilon = (i\varepsilon)^2\int a_2(y)e^{i\phi(y)/\varepsilon}dy. \tag{2.49}$$

where $a_2(y) = \text{div}(a_1(y)v(y))$. This then leads to the general recursion:

$$a_{m+1}(y) = \text{div}(a_m(y)v(y)) \implies I_\varepsilon = (i\varepsilon)^m\int a_m(y)e^{i\phi(y)/\varepsilon}dy. \tag{2.50}$$

which means the process can be repeated m times, each time gaining a factor of ε.

4 *Step 4: The Decay Estimate:* We have the assumptions such that: $a \in C_c^m(\mathbb{R}^n)$ (compactly supported, m-times differentiable), and $\phi \in C^{m+1}(\mathbb{R}^n)$ with $\nabla\phi(y) > 0$:

$$|I_\varepsilon| = \left|\int_{\mathbb{R}^n} a(y)e^{i/\varepsilon\phi(y)}dy\right| \leq \varepsilon^m\int_{\mathbb{R}^n}|a_m(y)|dy = \varepsilon^m\|a_m\|_{L^1(\mathbb{R}^n)}. \tag{2.51}$$

This holds for any $m \geq 1$:

$$|I_\varepsilon| \leq \varepsilon^m \|a_m\|_{L^1(\mathbb{R}^n)} \leq C_m \varepsilon^m. \tag{2.52}$$

where C_m depends on $\|D^\alpha a\|_{L^\infty}$ for $|\alpha| \leq m$, the derivatives of ϕ up to order $m+1$, and $\inf|\nabla\phi|^{-1}$ on $\text{supp}(a)$.

This argument proves **Lemma 1** (2.2) and formalizes the intuition from the **Fast Oscillation Path** from the **Fig 4: Logical flow of the stationary phase method**: that away from critical points, the relentless oscillations of $e^{i\phi(y)/\varepsilon}$ cause such effective self-cancellation that the integral's decay is faster than any polynomial power of ε ☺.

2.4.2 *Case 2: Stationary Phase.*

We now analyze the integral (2.13) near its critical points, which are the only regions contributing to the leading-order asymptotics; all other regions are $O(\varepsilon^m)$. Here, we distinguish two cases.

[Case 2-i: Stationary Phase ($\nabla\phi = 0$), Purely Quadratic Phase] Fundamental case:

$$\phi(y) = \frac{1}{2} y^T A y. \tag{2.53}$$

where A is real, symmetric, and invertible ($\det A \neq 0$), ensuring a unique, non-degenerate critical point at $y_0 = 0$ with Hessian A. Thus, A is diagonalizable:

$$A = \text{diag}(\lambda_1, \ldots, \lambda_n), \quad \text{with } \lambda_i \neq 0 \text{ for all } i = 1, \ldots, n. \tag{2.54}$$

The *signature* of A is defined as:

$$\text{sgn}(A) = (\text{number of } \lambda_i > 0) - (\text{number of } \lambda_i < 0). \tag{2.55}$$

[Lemma 2 (Quadratic Phase)] For $\phi(y) = \frac{1}{2} y^T A y$ (A symmetric, $\det A \neq 0$):

$$\int_{\mathbb{R}^n} e^{\frac{i}{2\varepsilon} y^T A y} a(y) dy = \frac{(2\pi\varepsilon)^{n/2}}{|\det A|^{1/2}} e^{\frac{i\pi}{4}\text{sgn}(A)} (a(0) + O(\varepsilon)). \tag{2.56}$$

Proof: Exact Gaussian integral via diagonalization.

The objective for **Case 2-i** is to prove **Lemma 2** (2.2).

Proof. The integral:

$$I_\varepsilon = \int_{\mathbb{R}^n} a(y) e^{\frac{i}{2\varepsilon} y^T A y} dy. \tag{2.57}$$

is highly oscillatory and does not converge absolutely. We proceed in several steps.

1. *Step 1: Regularization via Tempered Distributions:*

As we know, A is a real symmetric matrix, $y^T Ay$ is a quadratic form, and $f(y) = e^{\frac{i}{2\varepsilon}y^T Ay}$ is a bounded function (since $|e^{i\theta}| = 1$). We recall that any bounded measurable function defines a tempered distribution $T_f \in \mathcal{S}'(\mathbb{R}^n)$ by the rule[5]:

$$\langle T_f, \phi \rangle = \int_{\mathbb{R}^n} f(y)\phi(y)dy, \quad \text{for } \phi(y) \in \mathcal{S}(\mathbb{R}^n). \tag{2.58}$$

[6] Since $\phi(y)$ is rapidly decreasing, this integral converges. This gives us:

$$T_f \in \mathcal{S}'(\mathbb{R}^n). \tag{2.59}$$

[7]Moreover, because $\phi \in \mathcal{S}(\mathbb{R}^n)$ implies $\phi \in L^1(\mathbb{R}^n)$, the integral converges absolutely:

$$\left| \int_{\mathbb{R}^n} e^{\frac{i}{2\varepsilon}y^T Ay}\phi(y)dy \right| \leq \int_{\mathbb{R}^n} \underbrace{\left| e^{\frac{i}{2\varepsilon}y^T Ay} \right|}_{=1} |\phi(y)|dy = \|\phi\|_{L^1} < \infty. \tag{2.60}$$

This framework enables rigorous treatment of the high-frequency ($\varepsilon \to 0$) asymptotics through distributional Fourier analysis, which is essential for the stationary phase method.

2. *Step 2: Distributional Duality:* The key to evaluating $I_\varepsilon = \langle T_f, a \rangle$ is the following duality property of the Fourier transform on $\mathcal{S}'(\mathbb{R}^n)$. For a tempered distribution T_f and a Schwartz function $a \in \mathcal{S}(\mathbb{R}^n)$, the following identity holds [8]:

$$\int_{\mathbb{R}^n} a(y)f(y)dy = \int_{\mathbb{R}^n} \widehat{a}(-\xi)\,\widehat{f}(\xi)d\xi. \tag{2.61}$$

[5]A **tempered distribution** T is a bounded linear functional on the Schwartz space $\mathcal{S}(\mathbb{R}^n)$. That is, $T : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ is linear, and if a sequence of Schwartz functions $f_n \to 0$ in $\mathcal{S}(\mathbb{R}^n)$, then $T(f_n) \to 0$ in $\mathbb{C}$.

- **Definition (Fourier Transform):** For any $T \in \mathcal{S}'(\mathbb{R}^n)$, its **Fourier transform** $\widehat{T} \in \mathcal{S}'(\mathbb{R}^n)$ is defined by its action on test functions $f \in \mathcal{S}(\mathbb{R}^n)$:

  $$\widehat{T}(f) = T(\widehat{f}) \quad \text{or equivalently} \quad \langle \widehat{T}, f \rangle = \langle T, \widehat{f} \rangle.$$

  This definition generalizes the classical Fourier transform and is the foundation for the proof of Lemma 2. (If a distribution T is given by a function ϕ: $\forall \phi \in L^2(\mathbb{R}^n) : T(f) = \int_{\mathbb{R}^n} \phi(x)f(x)dx$, then this definition ensures $\widehat{T}$ is the distribution given by the usual Fourier transform $\widehat{\phi}$.)

- **Continuity:** A sequence $\{f_n\} \subset \mathcal{S}(\mathbb{R}^n)$ converges to 0 if for all multi-indices $\alpha, \beta \in \mathbb{N}_0^n$:

  $$\sup_{x \in \mathbb{R}^n} \left| x^\alpha \partial^\beta f_n(x) \right| \to 0 \quad \text{as } n \to \infty.$$

[6]The **Schwartz space** (or space of rapidly decreasing functions) on $\mathbb{R}^n$ is defined as:

$$\mathcal{S}(\mathbb{R}^n) = \left\{ f \in C^\infty(\mathbb{R}^n) : \sup_{x \in \mathbb{R}^n} \left| x^\alpha \partial^\beta f(x) \right| < \infty \quad \text{for all multi-indices } \alpha, \beta \in \mathbb{N}^n \right\}.$$

A sequence $\{f_n\}$ converges to 0 in $\mathcal{S}(\mathbb{R}^n)$ if $\sup_{x \in \mathbb{R}^n} \left| x^\alpha \partial^\beta f_n(x) \right| \to 0$ for all α, β. This space is closed under the operations of differentiation (∂^γ) and multiplication by polynomials (x^γ) for any multi-index γ.

[7]$\mathcal{S}'(\mathbb{R}^n)$: The **continuous dual of Schwartz space**. Contains functions of polynomial growth (L^p, bounded functions, etc.). Closed under Fourier transform, differentiation, and convolution.

where $\widehat{f}$ is the Fourier transform of f in the sense of distributions.

Applying this to our case, with $f(y) = e^{\frac{i}{2\varepsilon}y^T Ay}$, we obtain:

$$I_\varepsilon = \int_{\mathbb{R}^n} a(y) e^{\frac{i}{2\varepsilon}y^T Ay} dy = \int_{\mathbb{R}^n} \widehat{a}(-\xi)\, \mathcal{F}\left[e^{\frac{i}{2\varepsilon}y^T Ay}\right](\xi) d\xi. \tag{2.62}$$

The problem now reduces to computing the distributional Fourier transform $\mathcal{F}\left[e^{\frac{i}{2\varepsilon}y^T Ay}\right](\xi)$.

3. *Step 3: Regularization via a Damping Factor:* We need to compute $\hat{f}(\xi) = \mathcal{F}[e^{\frac{i}{2\varepsilon}y^T Ay}](\xi)$. The formal integral is not absolutely convergent because $|e^{i\phi(y)}| = 1$ (**Note:** In contrast, Eq. 2.60 converges absolutely because it also has the Schwartz function $\phi(y)$ providing decay). To solve this, we introduce a regularization parameter $e^{-\delta|y|^2}(\delta > 0)$ [9]:

$$\mathcal{F}_\delta(\xi) = \int_{\mathbb{R}^n} e^{\frac{i}{2\varepsilon}y^T Ay - \delta|y|^2 - iy\cdot\xi} dy. \tag{2.63}$$

The damping term ensures the integral converges absolutely. The distributional Fourier transform is then defined as the limit:

$$\mathcal{F}\left[e^{\frac{i}{2\varepsilon}y^T Ay}\right](\xi) = \lim_{\delta\to 0^+} \mathcal{F}_\delta(\xi). \tag{2.64}$$

Physically, this represents adding dissipation to the system and then removing it ($\delta \to 0$) to recover the original dynamics.

4. *Step 4: Diagonalization of the Regularized Integral:* We now compute $\mathcal{F}_\delta(\xi)$. Since A is symmetric, we decompose it as $A = Q^T DQ$, where $D = \mathrm{diag}(\lambda_1, \cdots, \lambda_n)$. The change of variables $y = Qz$ yields:

$$y^T Ay = (Qz)^T A(Qz) = z^T Q^T AQz = z^T Dz = \sum_{j=1}^n \lambda_j z_j^2. \tag{2.65}$$

The integral now factorizes into a product of 1D integrals:

$$I_\varepsilon = \int_{\mathbb{R}^n} \widehat{a}(-\xi) \left(\prod_{j=1}^n \mathcal{F}[e^{i\lambda_j z_j^2/\varepsilon}](\xi_j)\right) d\xi. \tag{2.66}$$

This decouples the multidimensional problem into independent 1D Gaussians.

5. *Step 5: The One-Dimensional Computation:*

   5.1 **1D Fourier Transform (Preliminary):** For $n = 1$, we compute $\mathcal{F}[e^{i\lambda x^2}](\xi)$ using regularization:

$$\mathcal{F}[e^{i\lambda x^2}](\xi) = \lim_{\delta\to 0^+} \int_{-\infty}^{\infty} e^{i\lambda x^2 - \delta x^2 - ix\xi} dx. \tag{2.67}$$

[8] This is the **distributional version of Plancherel's theorem**. The classical L^2 theorem is a special case of this more general duality, which is the true foundation for the Fourier analysis of distributions.

[9] Recall our convention for the Fourier transform: $\mathcal{F}[f](\xi) = \int f(x)e^{-ix\cdot\xi} dx$. This is the convention used in standard PDE theory.

This is a Gaussian integral with complex coefficients. It is solvable via completing the square.

5.2 **Completing the Square for the Regularized Fourier Transform:**

Now, we want to compute the 1D Fourier transform of the regularized oscillatory exponential (2.67). We can do the following steps:

5.2.1 **Combine and Rewrite the Exponent:** We start with the exponent in the integrand:

$$i\lambda x^2 - \delta x^2 - ix\xi = (i\lambda - \delta)x^2 - ix\xi = (i\lambda - \delta)x^2 - ix\xi = -\alpha x^2 - ix\xi. \tag{2.68}$$

with $\alpha = \delta - i\lambda$ ($\mathrm{Re}(\alpha) = \delta > 0$).

5.2.2 **Complete the Square:** Rewrite the exponent as the below:

$$-\alpha x^2 - ix\xi = -\alpha\left(x^2 + \frac{i\xi}{\alpha}x\right) \implies x^2 + \frac{i\xi}{\alpha}x = \left(x + \frac{i\xi}{2\alpha}\right)^2 - \left(\frac{i\xi}{2\alpha}\right)^2. \tag{2.69}$$

Thus, we have:

$$-\alpha\left[\left(x + \frac{i\xi}{2\alpha}\right)^2 - \frac{\xi^2}{4\alpha^2}\right] = -\alpha\left(x + \frac{i\xi}{2\alpha}\right)^2 + \frac{\xi^2}{4\alpha}. \tag{2.70}$$

5.2.3 **Evaluate the Gaussian Integral:** Substitute back into the integral:

$$\int_{-\infty}^{\infty} e^{-\alpha\left(x+\frac{i\xi}{2\alpha}\right)^2 + \frac{\xi^2}{4\alpha}} dx = e^{\frac{\xi^2}{4\alpha}} \int_{-\infty}^{\infty} e^{-\alpha\left(x+\frac{i\xi}{2\alpha}\right)^2} dx. \tag{2.71}$$

Let $y = x + \frac{i\xi}{2\alpha}$ (note that $dy = dx$ since the shift is constant in x) [10]:

$$e^{\frac{\xi^2}{4\alpha}} \int_{-\infty}^{\infty} e^{-\alpha y^2} dy = e^{\frac{\xi^2}{4\alpha}} \sqrt{\frac{\pi}{\alpha}}. \tag{2.72}$$

5.3 **Taking the Limit $\delta \to 0^+$:**

5.3.1 **Simplify and Take the Limit:** Substitute $\alpha = \delta - i\lambda$:

$$\int_{-\infty}^{\infty} e^{i\lambda x^2 - \delta x^2 - ix\xi} dx = e^{\frac{\xi^2}{4\alpha}} \sqrt{\frac{\pi}{\alpha}} = e^{\frac{\xi^2}{4(\delta - i\lambda)}} \sqrt{\frac{\pi}{\delta - i\lambda}}. \tag{2.73}$$

Now take $\delta \to 0^+$:

$$\mathcal{F}[e^{i\lambda x^2}](\xi) = \lim_{\delta\to 0^+} \int_{-\infty}^{\infty} e^{i\lambda x^2 - \delta x^2 - ix\xi} dx = \lim_{\delta\to 0^+} e^{\frac{\xi^2}{4(\delta - i\lambda)}} \sqrt{\frac{\pi}{\delta - i\lambda}} = \sqrt{\frac{\pi}{-i\lambda}} e^{-\frac{i\xi^2}{4\lambda}}. \tag{2.74}$$

The damping factor $e^{-\delta x^2}$ ensures convergence for $\delta > 0$, and the limit $\delta \to 0^+$ is well-defined. The phase factor term $e^{-\frac{i\xi^2}{4\lambda}}$ encodes the oscillation in frequency

[10] The standard Gaussian integral $\int_{-\infty}^{\infty} e^{-\alpha y^2} dy = \sqrt{\frac{\pi}{\alpha}}$ for $\mathrm{Re}(\alpha) > 0$.

space, linked to the stationary phase at $x = \frac{\xi}{2\lambda}$ [11].

5.3.2 **Handling the Branch Cut:** The term can be considered as:

$$\sqrt{-i\lambda} = \sqrt{|\lambda|}e^{-i\frac{\pi}{4}\text{sgn}(\lambda)}. \tag{2.75}$$

This ensures the consistency with the signature of λ (positive/negative eigenvalues in n dimensions.

5.3.2.1 **Branch cut in Complex Square Root:** The square root of a complex number $z = re^{i\theta}$ (where $r > 0$ and $\theta \in (-\pi, \pi]$) is:

$$\sqrt{z} = \sqrt{r}e^{i\theta/2}. \tag{2.76}$$

Typically, we place the branch cut along the negative real axis, meaning θ is restricted to $(-\pi, \pi]$.

5.3.2.2 **Evaluating $\sqrt{-i\lambda}$:** Let $\lambda \in \mathbb{R}\backslash\{0\}$. We analyze two cases:

Case i: $\lambda > 0$: Since $-i = e^{-i\pi/2}$, we can write $-i\lambda$ in polar form as:

$$-i\lambda = \lambda e^{-i\pi/2} \implies \sqrt{-i\lambda} = \sqrt{\lambda}e^{-i\pi/4} = |\lambda|^{1/2}e^{-i\pi/4}. \tag{2.77}$$

Case ii: $\lambda < 0$: Since $\lambda < 0$ implies $-i\lambda = |\lambda|i = |\lambda|e^{i\pi/2}$, we can have:

$$-i\lambda = |\lambda|e^{i\pi/2} \implies \sqrt{-i\lambda} = \sqrt{|\lambda|}e^{i\pi/4} = |\lambda|^{1/2}e^{i\pi/4}. \tag{2.78}$$

Combine the results by using the signum function $\text{sgn}(\lambda)$, we have:

$$\sqrt{-i\lambda} = |\lambda|^{-1/2}e^{-i\frac{\pi}{4}\text{sgn}(\lambda)}. \tag{2.79}$$

So, for $\lambda > 0 : \text{sgn}(\lambda) = +1$ gives $e^{-i\pi/4}$, and for $\lambda < 0 : \text{sgn}(\lambda) = -1$ gives $e^{i\pi/4}$. The term $e^{-i\frac{\pi}{4}\text{sgn}(\lambda)}$ ensures the correct phase continuity across $\lambda = 0$. Moreover, in the n-dimensional case, this generalizes to the signature $\text{sgn}(A)$ (number of positive/negative eigenvalues).

5.3.2.3 **The final result:** The above analysis helps us to transfer (2.74) as the below expression:

$$\mathcal{F}[e^{i\lambda x^2}](\xi) = \sqrt{\frac{\pi}{-i\lambda}}e^{-\frac{i\xi^2}{4\lambda}} = \sqrt{\frac{\pi}{|\lambda|}}e^{i\frac{\pi}{4}\text{sgn}(\lambda)-\frac{i\xi^2}{4\lambda}}. \tag{2.80}$$

After the transformation, we noticed that the Fourier transform of $e^{i\lambda x^2}$localizes in frequency space, with amplitude decay $|\lambda|^{-1/2}$. This 1D result generalizes to higher dimensions via diagonalization, which can be shown in the next step.

[11]The integral $\int e^{i\lambda x^2 - ix\xi}dx$ is dominated by points where the phase $\phi(x) = \lambda x^2 - x\xi$ is stationary (e.g., where $\phi(x) = 0$): $\phi'(x) = 2\lambda x - \xi = 0 \implies x = \frac{\xi}{2\lambda}$. The phase $\phi(x)$varies slowly, so oscillations cancel out less destructively.

6. *Step 6: Multidimensional Case:* We now generalize the 1D result from Step 5 to prove **Lemma 2**, (2.2) in $\mathbb{R}^n$. The foundational 1D calculation is summarized in the following claim:

> **Claim (Fourier Transform of Pure Phase).** For any $\lambda \neq 0$, the tempered distribution $e^{i\lambda x^2} \in \mathcal{S}'(\mathbb{R})$ satisfies:
>
> $$\mathcal{F}[e^{i\lambda x^2}](\xi) = \lim_{\delta \to 0^+} \frac{\sqrt{\pi}}{\sqrt{\delta - i\lambda}} e^{\xi^2/(4(i\lambda - \delta))} \quad \text{(Regularized)} \tag{2.81}$$
>
> $$= \sqrt{\frac{\pi}{-i\lambda}} e^{-i\xi^2/(4\lambda)} \quad \text{(Distributional Limit)} \tag{2.82}$$

6.1 **Fourier Transform of $e^{i\lambda x^2}$ as a Tempered Distribution:**

6.1.1 **Tempered Distribution:** Define the functional $f_\lambda : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$, for $f_\lambda(x) = e^{i\lambda|x|^2}$ ($\lambda \neq 0$) by its action on Schwartz functions:

$$\langle f_\lambda, \phi \rangle = \int_{\mathbb{R}^n} e^{i\lambda|x|^2} \phi(x)\, dx, \quad \phi \in \mathcal{S}(\mathbb{R}^n). \tag{2.83}$$

Its Fourier transform $\mathcal{F}(f_\lambda)$ acts on test functions via:

$$\langle \mathcal{F}(f_\lambda), \phi \rangle = \langle f_\lambda, \widehat{\phi} \rangle = \int_{\mathbb{R}^n} e^{i\lambda|x|^2} \widehat{\phi}(x)\, dx. \tag{2.84}$$

where $\widehat{\phi}(x) = \int_{\mathbb{R}^n} \phi(\xi) e^{-ix\cdot\xi}\, d\xi$ is the Fourier transform of ϕ. [12]

6.1.2 **Introduce the Damping Factor:** Regularize f_λ with $e^{-\delta|x|^2}$ ($\delta > 0$):

$$\langle f_\lambda^\delta, \phi \rangle = \int_{\mathbb{R}^n} e^{i\lambda|x|^2 - \delta|x|^2} \phi(x) dx = \int_{\mathbb{R}^n} e^{-(\delta - i\lambda)|x|^2} \phi(x) dx. \tag{2.85}$$

6.1.3 **Dominated Convergence Theorem:** We now show that $f_\lambda^\delta \to f_\lambda$ in $\mathcal{S}'$ as $\delta \to 0^+$. Recall that convergence in $\mathcal{S}'$ is defined by convergence for every test function. The Fourier transform of a distribution is defined by $\langle \mathcal{F}(f_\lambda), \phi \rangle = \langle f_\lambda, \widehat{\phi} \rangle$. Therefore, to use the DCT, we must consider the specific test function $\psi = \widehat{\phi}$ in the pairing:

$$\langle f_\lambda^\delta, \widehat{\phi} \rangle = \int_{\mathbb{R}^n} e^{i\lambda|x|^2 - \delta|x|^2} \widehat{\phi}(x) dx. \tag{2.86}$$

We will apply the DCT to this integral. For any fixed $\phi \in \mathcal{S}(\mathbb{R}^n)$:

i. **Pointwise Convergence:** The integrand converges pointwise:

$$e^{i\lambda|x|^2 - \delta|x|^2} \widehat{\phi}(x) \to e^{i\lambda|x|^2} \widehat{\phi}(x) \quad \text{as } \delta \to 0^+ \text{ for all } x \in \mathbb{R}^n. \tag{2.87}$$

This holds because $e^{-\delta|x|^2} \to 1$ pointwise for all x.

[12] Some texts define the pairing with $\overline{\widehat{\phi}(x)}$ to align with the inverse Fourier transform or sesquilinear duality. Here, we follow Evans' convention for linearity.

ii. **Uniform Dominance by an L^1 Function:** For all $\delta \in (0,1]$ and all $x \in \mathbb{R}^n$, the integrand is bounded as follows:

$$\begin{aligned}\left|e^{i\lambda|x|^2-\delta|x|^2}\widehat{\phi}(x)\right| &= \left|e^{i\lambda|x|^2}\right| \cdot \left|e^{-\delta|x|^2}\right| \cdot \left|\widehat{\phi}(x)\right| && \text{(Absolute value)} \\ &= 1 \cdot e^{-\delta|x|^2} \cdot \left|\widehat{\phi}(x)\right| && \text{(Since } |e^{i\theta}| = 1\text{)} \\ &\leq 1 \cdot \left|\widehat{\phi}(x)\right| = \left|\widehat{\phi}(x)\right| && \text{(Since } 0 < e^{-\delta|x|^2} \leq 1\text{)} \end{aligned} \tag{2.88}$$

Since $\phi \in \mathcal{S}(\mathbb{R}^n)$, its Fourier transform $\widehat{\phi}$ is also in $\mathcal{S}(\mathbb{R}^n)$. By the definition of $\mathcal{S}(\mathbb{R}^n)$, for any integer $N > 0$, the function $(1+|x|^2)^N\widehat{\phi}(x)$ is bounded. That is, there exists a constant $C_N > 0$ such that:

$$(1+|x|^2)^N|\widehat{\phi}(x)| \leq C_N \quad \text{for all } x \in \mathbb{R}^n. \tag{2.89}$$

Dividing both sides by the positive function $(1+|x|^2)^N$ yields:

$$|\widehat{\phi}(x)| \leq C_N(1+|x|^2)^{-N} \quad \text{for all } x \in \mathbb{R}^n. \tag{2.90}$$

We choose $N = n+1$. The function $G(x) = C_{n+1}(1+|x|^2)^{-(n+1)}$ is integrable on $\mathbb{R}^n$ because the exponent $n+1 > n/2$[13]. Therefore, the original integrand is dominated by an L^1 function independent of δ.

By the Dominated Convergence Theorem, we may take the limit inside the integral:

$$\begin{aligned}\lim_{\delta\to 0^+}\langle f_\lambda^\delta, \widehat{\phi}\rangle &= \lim_{\delta\to 0^+}\int_{\mathbb{R}^n} e^{i\lambda|x|^2-\delta|x|^2}\widehat{\phi}(x)dx \\ &= \int_{\mathbb{R}^n}\lim_{\delta\to 0^+}\left(e^{i\lambda|x|^2-\delta|x|^2}\widehat{\phi}(x)\right)dx \\ &= \int_{\mathbb{R}^n} e^{i\lambda|x|^2}\widehat{\phi}(x)dx = \langle f_\lambda, \widehat{\phi}\rangle. \end{aligned} \tag{2.91}$$

Since this holds for all $\phi \in \mathcal{S}(\mathbb{R}^n)$, we conclude that $f_\lambda^\delta \to f_\lambda$ in $\mathcal{S}'(\mathbb{R}^n)$.

6.2 **Regularized Fourier Transform Calculation:** This step provides a self-contained, detailed proof for the **Claim** stated above. For completeness and to make this section fully independent, we repeat the core 1D calculation that was motivated in Steps 1–5, formalizing the result for the one-dimensional case.

6.2.1 **Integral Setup:** For $\delta > 0$ and $\lambda \neq 0$, compute:

$$\int_{\mathbb{R}} e^{i\lambda x^2-\delta x^2-ix\xi}\,dx. \tag{2.92}$$

[13]To prove $G(x) = C(1+|x|^2)^{-(n+1)} \in L^1(\mathbb{R}^n)$, switch to spherical coordinates:

$$\int_{\mathbb{R}^n} G(x)dx = \omega_{n-1}\int_0^\infty \frac{r^{n-1}}{(1+r^2)^{n+1}}dr.$$

The integral converges if and only if the exponent $N = n+1 > n/2$.

where the integrand is the regularized version of $e^{i\lambda x^2}$.

6.2.2 **Completing the Square:** Let $\alpha := \delta - i\lambda$ ($\mathrm{Re}(\alpha) = \delta > 0$). The exponent becomes:

$$(i\lambda - \delta)x^2 - ix\xi = -\alpha x^2 - ix\xi = -\alpha\left(x + \frac{i\xi}{2\alpha}\right)^2 - \frac{\xi^2}{4\alpha}. \tag{2.93}$$

Thus:

$$e^{i\lambda x^2 - \delta x^2 - ix\xi} = e^{-\xi^2/(4\alpha)} \cdot e^{-\alpha(x + i\xi/(2\alpha))^2}. \tag{2.94}$$

6.2.3 **Contour Integration:** Change variables via $z = \sqrt{\alpha}\left(x + \frac{i\xi}{2\alpha}\right)$ ($\mathrm{Re}(\sqrt{\alpha}) > 0$):

$$\int_{\mathbb{R}} e^{-\alpha(x + i\xi/(2\alpha))^2}\, dx = \frac{1}{\sqrt{\alpha}} \int_{\Gamma} e^{-z^2}\, dz = \frac{\sqrt{\pi}}{\sqrt{\alpha}}. \tag{2.95}$$

where Γ is the line $z = \sqrt{\alpha}x + \frac{i\xi}{2\sqrt{\alpha}}$.

6.2.4 **Final Regularized Result:** Combining terms:

$$\int_{\mathbb{R}} e^{i\lambda x^2 - \delta x^2 - ix\xi}\, dx = \frac{\sqrt{\pi}}{(\delta - i\lambda)^{1/2}} e^{-\xi^2/(4(\delta - i\lambda))} = \frac{\sqrt{\pi}}{(\delta - i\lambda)^{1/2}} e^{\xi^2/(4(i\lambda - \delta))}. \tag{2.96}$$

Note: This matches our **Claim** for the regularized result.

6.3 **Generalization to $\mathbb{R}^n$**

6.3.1 **Integral Setup:** For a symmetric matrix $A = Q^T D Q$ ($D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$), consider:

$$f(x) = e^{\frac{i}{\varepsilon} x^T A x - \delta |x|^2}, \quad I(\xi) = \int_{\mathbb{R}^n} e^{\frac{i}{\varepsilon} x^T A x - \delta |x|^2 - i\xi \cdot x} dx. \tag{2.97}$$

6.3.2 **Diagonalization:** Change variables $y = Qx$ (Q orthogonal). Since Q is orthogonal, $x = Q^T y$, $dx = dy$, and $\xi \cdot x = \xi \cdot (Q^T y) = (Q\xi) \cdot y$:

$$x^T A x = y^T D y = \sum_{j=1}^{n} \lambda_j y_j^2. \tag{2.98}$$

Substituting into the integral yields:

$$I(\xi) = \int_{\mathbb{R}^n} \exp\left(\frac{i}{\varepsilon} \sum_{j=1}^{n} \lambda_j y_j^2 - \delta \sum_{j=1}^{n} y_j^2 - i \sum_{j=1}^{n} (Q\xi)_j y_j\right) dy = \prod_{j=1}^{n} \int_{\mathbb{R}} \exp\left(\left(\frac{i\lambda_j}{\varepsilon} - \delta\right) y_j^2 - i\xi_j y_j\right) dy_j. \tag{2.99}$$

where $\xi_j \equiv (Q\xi)_j$ denotes the j-th component of the vector ξ in the diagonalized coordinate system. The integral factorizes because the exponential is a product of functions each depending on a single variable y_j.

6.3.3 **Reduction to the One-Dimensional Case:** Each integral in the product is a one-dimensional

integral of the form solved in the **Claim** (Step 6.2) and in detail in **6.2.1-6.2.4**. For a fixed dimension j, we identify:

$$\text{Parameter: } \lambda \to \lambda_j/\varepsilon, \quad \text{Frequency: } \xi \to \xi_j \tag{2.100}$$

The integral for the j-th dimension is:

$$I_j = \int_{\mathbb{R}} e^{i(\lambda_j/\varepsilon)y_j^2 - \delta y_j^2 - iy_j\xi_j} dy_j. \tag{2.101}$$

6.3.4 **Application of the 1D Result:** Applying the formula from the **Claim** directly to I_j gives the solution for one dimension:

$$I_j = \frac{\sqrt{\pi}}{\left(\delta - i(\lambda_j/\varepsilon)\right)^{1/2}} \exp\left(\frac{\xi_j^2}{4(i(\lambda_j/\varepsilon) - \delta)}\right) = \frac{\sqrt{\pi}}{\sqrt{\delta - \frac{i\lambda_j}{\varepsilon}}} \exp\left(\frac{\xi_j^2}{4\left(\frac{i\lambda_j}{\varepsilon} - \delta\right)}\right). \tag{2.102}$$

6.3.5 **Final Regularized Result:** The full n-dimensional integral $I(\xi)$ is the product of the results from all n dimensions:

$$\begin{aligned} I(\xi) &= \prod_{j=1}^{n} I_j = \prod_{j=1}^{n} \left[\frac{\sqrt{\pi}}{\sqrt{\delta - \frac{i\lambda_j}{\varepsilon}}} \exp\left(\frac{\xi_j^2}{4\left(\frac{i\lambda_j}{\varepsilon} - \delta\right)}\right) \right] \\ &= \left(\prod_{j=1}^{n} \sqrt{\pi}\right) \cdot \left(\prod_{j=1}^{n} \frac{1}{\sqrt{\delta - \frac{i\lambda_j}{\varepsilon}}}\right) \cdot \left(\prod_{j=1}^{n} \exp\left(\frac{\xi_j^2}{4\left(\frac{i\lambda_j}{\varepsilon} - \delta\right)}\right)\right) \\ &= \pi^{n/2} \prod_{j=1}^{n} \frac{1}{\sqrt{\delta - \frac{i\lambda_j}{\varepsilon}}} \exp\left(\sum_{j=1}^{n} \frac{\xi_j^2}{4\left(\frac{i\lambda_j}{\varepsilon} - \delta\right)}\right). \end{aligned} \tag{2.103}$$

Since the exponential of a sum is the product of exponentials, the final result is:

$$I(\xi) = \pi^{n/2} \prod_{j=1}^{n} \frac{1}{\sqrt{\delta - \frac{i\lambda_j}{\varepsilon}}} \exp\left(\frac{\xi_j^2}{4\left(\frac{i\lambda_j}{\varepsilon} - \delta\right)}\right). \tag{2.104}$$

6.4 Plancherel and Taylor Expansion: The Final Proof of Lemma 2

6.4.1 **Regularized Integral Setup**: Our goal is to compute the original oscillatory integral:

$$I(\varepsilon) = \int_{\mathbb{R}^n} a(x) e^{\frac{i}{2\varepsilon} x^T A x} dx. \tag{2.105}$$

To make this rigorous, we work with the regularized version and take the limit:

$$I(\varepsilon) = \lim_{\delta \to 0^+} I^\delta(\varepsilon), \quad \text{where} \quad I^\delta(\varepsilon) = \int_{\mathbb{R}^n} a(x) e^{\frac{i}{2\varepsilon} x^T A x - \delta |x|^2} dx. \tag{2.106}$$

The function $a \in C_c^\infty(\mathbb{R}^n)$ provides the smooth, compactly supported amplitude.

6.4.2 **Application of Plancherel's Theorem**[14]: We now apply Plancherel's Theorem to the regularized integral. Let us define:

$$f(x) = a(x), \quad g^\delta(x) = e^{\frac{i}{2\varepsilon}x^T Ax - \delta|x|^2}. \tag{2.107}$$

For our real-valued functions, the complex conjugate can be handled carefully. A more straightforward approach is to note:

$$I^\delta(\varepsilon) = \int_{\mathbb{R}^n} a(x) g^\delta(x) dx = \int_{\mathbb{R}^n} \widehat{a}(\xi)\, \widehat{g^\delta}(-\xi) d\xi = \int_{\mathbb{R}^n} \widehat{a}(\xi)\, \mathcal{F}[g^\delta](-\xi) d\xi. \tag{2.108}$$

From Step 6.3, we have already computed the Fourier transform of $g^\delta(x)$. The calculation in **6.3.1-6.3.5** was for the function $e^{\frac{i}{\varepsilon}x^T Ax - \delta|x|^2}$. Our function here has a factor of $1/2$ in the exponent. To connect them, we simply make the replacement $A \to \frac{1}{2}A$ (or $\lambda_j \to \lambda_j/2$) in the result from Step 6.3.

6.4.3 **Inserting the Result from Step 6.3**: Applying the result from Eq. (X) in Step 6.3.5 with $A \to \frac{1}{2}A$ gives:

$$\mathcal{F}[g^\delta](\xi) = \pi^{n/2} \prod_{j=1}^n \frac{1}{\sqrt{\delta - \frac{i\lambda_j}{2\varepsilon}}} \exp\left(\frac{\xi_j^2}{4\left(\frac{i\lambda_j}{2\varepsilon} - \delta\right)}\right). \tag{2.109}$$

Therefore, the integral becomes:

$$I^\delta(\varepsilon) = \pi^{n/2} \int_{\mathbb{R}^n} \widehat{a}(\xi) \left(\prod_{j=1}^n \frac{1}{\sqrt{\delta - \frac{i\lambda_j}{2\varepsilon}}} \exp\left(\frac{\xi_j^2}{4\left(\frac{i\lambda_j}{2\varepsilon} - \delta\right)}\right)\right) d\xi. \tag{2.110}$$

We now take the limit $\delta \to 0^+$. Using the Dominated Convergence Theorem (justified by the rapid decay of $\widehat{a}(\xi) \in \mathcal{S}(\mathbb{R}^n)$), we deduce:

$$\begin{aligned} I(\varepsilon) &= \lim_{\delta \to 0^+} I^\delta(\varepsilon) \\ &= \pi^{n/2} \int_{\mathbb{R}^n} \widehat{a}(\xi) \left(\prod_{j=1}^n \frac{1}{\sqrt{-\frac{i\lambda_j}{2\varepsilon}}} \exp\left(\frac{\xi_j^2}{4\left(\frac{i\lambda_j}{2\varepsilon}\right)}\right)\right) d\xi \\ &= \pi^{n/2} \prod_{j=1}^n \frac{1}{\sqrt{-\frac{i\lambda_j}{2\varepsilon}}} \int_{\mathbb{R}^n} \widehat{a}(\xi) \exp\left(-\frac{i\varepsilon}{2} \sum_{j=1}^n \frac{\xi_j^2}{\lambda_j}\right) d\xi. \end{aligned} \tag{2.111}$$

[14]Plancherel's Theorem states that for functions $f, g \in L^2(\mathbb{R}^n)$, the following equality holds: $\int_{\mathbb{R}^n} f(x)\, \overline{g(x)} dx = \int_{\mathbb{R}^n} \widehat{f}(\xi)\, \overline{\widehat{g}(\xi)} d\xi$.

Note that $\sum_{j=1}^{n} \frac{\xi_j^2}{\lambda_j} = \xi^T D^{-1} \xi$. Since $A = Q^T D Q$, its inverse is $A^{-1} = Q^T D^{-1} Q$, and thus $\xi^T D^{-1} \xi = (Q\xi)^T D^{-1} (Q\xi) = (Q\xi)^T (Q A^{-1} Q^T)(Q\xi) = \xi^T A^{-1} \xi$ because $Q^T Q = I$. Therefore:

$$I(\varepsilon) = \pi^{n/2} \prod_{j=1}^{n} \frac{1}{\sqrt{-\frac{i\lambda_j}{2\varepsilon}}} \int_{\mathbb{R}^n} \widehat{a}(\xi) \exp\left(-\frac{i\varepsilon}{2} \xi^T A^{-1} \xi\right) d\xi. \tag{2.112}$$

6.4.4 **Signature and Determinant**: We now simplify the prefactor. Note that:

$$\begin{aligned} \prod_{j=1}^{n} \frac{1}{\sqrt{-\frac{i\lambda_j}{2\varepsilon}}} &= \prod_{j=1}^{n} \sqrt{\frac{2\varepsilon}{-i\lambda_j}} = (2\varepsilon)^{n/2} \prod_{j=1}^{n} \frac{1}{\sqrt{-i\lambda_j}} \\ &= (2\varepsilon)^{n/2} \frac{1}{\sqrt{\prod_{j=1}^{n} (-i\lambda_j)}} = (2\varepsilon)^{n/2} \frac{1}{\sqrt{(-i)^n \det A}}. \end{aligned} \tag{2.113}$$

Now, $(-i)^n = (e^{-i\pi/2})^n = e^{-in\pi/2}$. The signature term comes from evaluating $\sqrt{(-i)^n} = (e^{-in\pi/2})^{1/2} = e^{-in\pi/4}$, which is not quite general. The correct, general approach is to handle each eigenvalue's phase separately, as in Evans:

$$\begin{aligned} \prod_{j=1}^{n} \frac{1}{\sqrt{-i\lambda_j}} &= \prod_{j=1}^{n} \frac{1}{\sqrt{|\lambda_j|}} \cdot \frac{1}{\sqrt{e^{-i\frac{\pi}{2}\operatorname{sgn}(\lambda_j)}}} = \frac{1}{|\det A|^{1/2}} \prod_{j=1}^{n} e^{i\frac{\pi}{4}\operatorname{sgn}(\lambda_j)} \\ &= \frac{e^{i\frac{\pi}{4}\sum_{j=1}^{n} \operatorname{sgn}(\lambda_j)}}{|\det A|^{1/2}} = \frac{e^{i\frac{\pi}{4}\operatorname{sgn}(A)}}{|\det A|^{1/2}}. \end{aligned} \tag{2.114}$$

Therefore, the prefactor simplifies to:

$$\pi^{n/2} \cdot (2\varepsilon)^{n/2} \cdot \frac{e^{i\frac{\pi}{4}\operatorname{sgn}(A)}}{|\det A|^{1/2}} = (2\pi\varepsilon)^{n/2} \frac{e^{i\frac{\pi}{4}\operatorname{sgn}(A)}}{|\det A|^{1/2}}. \tag{2.115}$$

Thus, we have arrived at the core expression:

$$I(\varepsilon) = (2\pi\varepsilon)^{n/2} \frac{e^{i\frac{\pi}{4}\operatorname{sgn}(A)}}{|\det A|^{1/2}} \int_{\mathbb{R}^n} \widehat{a}(\xi) \exp\left(-\frac{i\varepsilon}{2} \xi^T A^{-1} \xi\right) d\xi. \tag{2.116}$$

6.4.5 **Taylor Expansion and Dominated Convergence**: To extract the asymptotic behavior as $\varepsilon \to 0$, we expand the exponential in the integrand:

$$\exp\left(-\frac{i\varepsilon}{2} \xi^T A^{-1} \xi\right) = 1 - \frac{i\varepsilon}{2} \xi^T A^{-1} \xi + O(\varepsilon^2 |\xi|^4). \tag{2.117}$$

Substituting this into the integral gives:

$$\int_{\mathbb{R}^n} \widehat{a}(\xi) \exp\left(-\frac{i\varepsilon}{2} \xi^T A^{-1} \xi\right) d\xi = \int_{\mathbb{R}^n} \widehat{a}(\xi) d\xi - \frac{i\varepsilon}{2} \int_{\mathbb{R}^n} \widehat{a}(\xi) \xi^T A^{-1} \xi d\xi + O(\varepsilon^2) \tag{2.118}$$

$$= (2\pi)^{n/2} a(0) + O(\varepsilon). \tag{2.119}$$

The leading term uses the Fourier inversion theorem: $\int \widehat{a}(\xi)d\xi = (2\pi)^{n/2}a(0)$. The error term is $O(\varepsilon)$ because $\widehat{a}(\xi)$ is a Schwartz function, ensuring $\int |\widehat{a}(\xi)||\xi|^2 d\xi < \infty$.

6.4.6 **Final Result**: Combining the results from steps 6.4.4 and 6.4.5, we obtain the desired asymptotic formula:

$$\int_{\mathbb{R}^n} a(x)e^{\frac{i}{2\varepsilon}x^T Ax}dx = (2\pi\varepsilon)^{n/2}\frac{e^{i\frac{\pi}{4}\mathrm{sgn}(A)}}{|\det A|^{1/2}}\left[a(0)+O(\varepsilon)\right] \quad \text{as } \varepsilon \to 0. \tag{2.120}$$

This completes the proof of **Lemma 2**, 2.2.

□

Moreover, the above also shows the **Stationary Phase Dominant Path** from the **Fig 4: Logical flow of the stationary phase method** without using **Lemma 3(ii)**. For the complete version with using **Morse Lemma**, we can see the below case.

[Case 2-ii: Stationary Phase ($\nabla\phi = 0$, $\det\left(D^2\phi(y_0)\right) \neq 0$), General Non-Degenerate Critical Point] We now consider the general case where the phase ϕ has a non-degenerate critical point at some y_0:

$$\nabla\phi(y_0) = 0, \quad \det\left(D^2\phi(y_0)\right) \neq 0. \tag{2.121}$$

The phase is not necessarily quadratic, but the Morse Lemma (Lemma 3(ii)) guarantees we can make it so by a change of coordinates.

This is the heart of the stationary phase method for a single non-degenerate critical point.

Proof. 1. *Step 1: Localize with a Partition of Unity.* As before, we use a partition of unity $\{\psi_k\}$ to isolate a small neighborhood $B(y_0, \delta)$ around the critical point. We analyze the local integral:

$$I_\varepsilon^{(y_0)} = \int_{B(y_0,\delta)} \psi(y)a(y)e^{i\phi(y)/\varepsilon}dy. \tag{2.122}$$

The function $\psi(y)$ is a bump function that is 1 on $B(y_0, \delta/2)$ and 0 outside $B(y_0, \delta)$, ensuring we only study the region near y_0.

2. *Step 2: Apply the Morse Lemma (Lemma 3(ii)).* The Morse Lemma provides a smooth change of variables $y = \mathbf{\Phi}(x)$ defined near y_0 such that:

   i. $\mathbf{\Phi}(0) = y_0$

   ii. $D\mathbf{\Phi}(0) = I$ (so the transformation is volume-preserving to first order)

   iii. $\phi(\mathbf{\Phi}(x)) = \phi(y_0) + \frac{1}{2}x^T Ax$, where $A = D^2\phi(y_0)$

Perform this change of variables in the integral:

$$I_\varepsilon^{(y_0)} = \int_{\mathbf{\Phi}^{-1}(B(y_0,\delta))} \psi(\mathbf{\Phi}(x))\, a(\mathbf{\Phi}(x))\, e^{i\left(\phi(y_0)+\frac{1}{2}x^T Ax\right)/\varepsilon}\, |\det D\mathbf{\Phi}(x)|\, dx$$
$$= e^{i\phi(y_0)/\varepsilon} \int_{\mathbf{\Phi}^{-1}(B(y_0,\delta))} \underbrace{\psi(\mathbf{\Phi}(x))\, a(\mathbf{\Phi}(x))\, |\det D\mathbf{\Phi}(x)|}_{\text{New amplitude } \tilde{a}(x)}\, e^{\frac{i}{2\varepsilon}x^T Ax} dx. \tag{2.123}$$

3. *Step 3: Apply Lemma 2 to the Transformed Integral.* The integral is now in the canonical form of **Case 2-i**. The phase is purely quadratic: $\frac{1}{2}x^T Ax$. The new amplitude is $\tilde{a}(x)$. We can directly apply Lemma 2:

$$I_\varepsilon^{(y_0)} = e^{i\phi(y_0)/\varepsilon}\left[\frac{(2\pi\varepsilon)^{n/2}}{|\det A|^{1/2}} e^{\frac{i\pi}{4}\mathrm{sgn}(A)}\left(\tilde{a}(0)+O(\varepsilon)\right)\right]$$
$$= \frac{(2\pi\varepsilon)^{n/2}}{|\det D^2\phi(y_0)|^{1/2}} e^{\frac{i\pi}{4}\mathrm{sgn}(D^2\phi(y_0))} e^{i\phi(y_0)/\varepsilon}\left(\tilde{a}(0)+O(\varepsilon)\right). \tag{2.124}$$

4. *Step 4: Interpret the Result.* Evaluate $\tilde{a}(0)$:

$$\tilde{a}(0) = \psi(\mathbf{\Phi}(0))\, a(\mathbf{\Phi}(0))\, |\det D\mathbf{\Phi}(0)| = \psi(y_0)\, a(y_0)\, |\det I| = a(y_0). \tag{2.125}$$

The last equality holds because $\psi(y_0) = 1$ since y_0 is in the support of the bump function. Therefore, the contribution from this critical point is:

$$I_\varepsilon^{(y_0)} = \frac{(2\pi\varepsilon)^{n/2}}{|\det D^2\phi(y_0)|^{1/2}} e^{\frac{i\pi}{4}\mathrm{sgn}(D^2\phi(y_0))} e^{i\phi(y_0)/\varepsilon}\left(a(y_0)+O(\varepsilon)\right). \tag{2.126}$$

□

This shows the **Stationary Phase Dominant Path** from the **Fig 4: Logical flow of the stationary phase method** completely ☺.

2.4.3 *Applying the Stationary Phase Formula to the Wave Equation*

We now apply the stationary phase method to the oscillatory integral representation of the wave solution derived earlier. Recall the expression:

$$I_{\pm}^{\varepsilon}(x,t) = \frac{1}{(2\pi\varepsilon)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(z) \exp\left(\frac{i}{\varepsilon}\phi_{\pm}(x,z,\xi,t)\right) dz d\xi. \tag{2.127}$$

where the phase function is given by:

$$\phi_{\pm}(x,z,\xi,t) = p(z) + \xi \cdot (x-z) \pm t|\xi|. \tag{2.128}$$

The full solution is a superposition of forward and backward waves:

$$u^{\varepsilon}(x,t) = \frac{1}{2}\left(I_{+}^{\varepsilon}(x,t) + I_{-}^{\varepsilon}(x,t)\right). \tag{2.129}$$

1. *Step 1: Find the Stationary Points:* The dominant contributions in the limit $\varepsilon \to 0$ come from points where the phase is stationary, i.e., where its gradient vanishes. We compute the critical points with respect to the integration variables z and ξ.

   1.1 **Compute the Gradient:** We calculate the partial derivatives of the phase function $\phi_{\pm}(x,z,\xi,t)$:

   $$\nabla_z \phi_{\pm} = \nabla_z(p(z) + \xi \cdot (x-z) \pm t|\xi|) = \nabla p(z) - \xi, \tag{2.130}$$
   $$\nabla_\xi \phi_{\pm} = \nabla_\xi(p(z) + \xi \cdot (x-z) \pm t|\xi|) = (x-z) \pm t\nabla_\xi|\xi|. \tag{2.131}$$

   Recall that for $\xi \neq 0$, $\nabla_\xi|\xi| = \frac{\xi}{|\xi|}$.

   1.2 **Set the Gradient to Zero:** Setting the derivatives to zero defines the stationary set $S^{\pm}$:

   $$\nabla_z \phi_{\pm} = 0 \quad \Longrightarrow \quad \xi_0 = \nabla p(z_0), \tag{2.132}$$
   $$\nabla_\xi \phi_{\pm} = 0 \quad \Longrightarrow \quad x - z_0 \pm t\frac{\xi_0}{|\xi_0|} = 0. \tag{2.133}$$

   1.3 **Solve the System:** Substituting the result from Equation (2.132) into Equation (2.133) yields the ray condition or eikonal equation:

   $$x = z_0 \mp t\frac{\nabla p(z_0)}{|\nabla p(z_0)|}. \tag{2.134}$$

   For a fixed observation point (x,t) and sufficiently small $t > 0$, the Implicit Function Theorem guarantees a unique solution $z_0 = z_0(x,t)$ to this equation. The corresponding stationary value for ξ is then $\xi_0 = \nabla p(z_0(x,t))$. This defines a unique ray path connecting the initial point z_0 to the observation point x at time t.

2. *Step 2: Compute the Hessian at the Stationary Point:* To apply the stationary phase formula, we need the Hessian matrix of second derivatives at the critical point (z_0, ξ_0).

2.1 **Define the Combined Variable:** Let $\mathbf{y} = (z, \xi)^T$ be the combined $2n$-dimensional variable of integration. The Hessian H is the $2n \times 2n$ matrix:

$$H = D^2_{\mathbf{y}}\phi_\pm = \begin{bmatrix} D^2_z\phi_\pm & D_z D_\xi \phi_\pm \\ D_\xi D_z \phi_\pm & D^2_\xi \phi_\pm \end{bmatrix}. \tag{2.135}$$

2.2 **Calculate the Second Derivatives:** We compute each block of the Hessian matrix:

i. $\nabla^2_z \phi_\pm = \nabla^2(p(z) - \xi \cdot z) = \nabla^2 p(z)$.

ii. $\nabla_z \nabla_\xi \phi_\pm = \nabla_z(\nabla p(z) - \xi) = -I$.

iii. $\nabla_\xi \nabla_z \phi_\pm = \nabla_\xi(x - z \pm t\frac{\xi}{|\xi|}) = -I$. (This ithe transpose of the previous block, confirming the symmetry of the Hessian).

iv. $\nabla^2_\xi \phi_\pm = \nabla_\xi \left(\pm t \frac{\xi}{|\xi|}\right) = \pm t \nabla_\xi \left(\frac{\xi}{|\xi|}\right)$.

The derivative $\nabla_\xi \left(\frac{\xi}{|\xi|}\right)$ is a matrix. Using the quotient rule or direct computation, for $\xi \neq 0$:

$$\nabla_\xi \left(\frac{\xi_i}{|\xi|}\right) = \frac{\delta_{ij}}{|\xi|} - \frac{\xi_i \xi_j}{|\xi|^3}. \tag{2.136}$$

Therefore, the matrix is $\nabla^2_\xi \phi_\pm = \pm t \left(\frac{I}{|\xi|} - \frac{\xi \otimes \xi}{|\xi|^3}\right)$.

2.3 **Assemble the Hessian:** Evaluating all blocks at the critical point (z_0, ξ_0) where $\xi_0 = \nabla p(z_0)$, we get the final form of the Hessian:

$$H = \begin{bmatrix} \nabla^2 p(z_0) & -I \\ -I & \pm t \left(\frac{I}{|\xi_0|} - \frac{\xi_0 \xi_0^T}{|\xi_0|^3}\right) \end{bmatrix}. \tag{2.137}$$

The matrix $P(\xi_0) = I - \frac{\xi_0 \xi_0^T}{|\xi_0|^2}$ is the projection matrix onto the space orthogonal to ξ_0. Note that $\frac{I}{|\xi_0|} - \frac{\xi_0 \xi_0^T}{|\xi_0|^3} = \frac{1}{|\xi_0|} P(\xi_0)$.

3. *Step 3: Apply the Stationary Phase Formula:* We now use the general stationary phase formula for an oscillatory integral in $\mathbb{R}^{2n}$.

3.1 **General Formula:** For an integral of the form $I = \int_{\mathbb{R}^{2n}} f(\mathbf{y}) e^{\frac{i}{\varepsilon}\phi(\mathbf{y})} d\mathbf{y}$ with a non-degenerate critical point at $\mathbf{y}_0$, the asymptotic expansion is:

$$I \sim \frac{(2\pi\varepsilon)^n}{|\det H|^{1/2}} e^{\frac{i}{\varepsilon}\phi(\mathbf{y}_0) + i\frac{\pi}{4}\mathrm{sgn}(H)} \left(f(\mathbf{y}_0) + O(\varepsilon)\right). \tag{2.138}$$

The factor $(2\pi\varepsilon)^n$ is the standard stationary phase constant for a $2n$D integral, and $\mathrm{sgn}(H)$ denotes the matrix signature (number of positive minus negative eigenvalues).

3.2 **Identify Our Functions:** In our case:

i. $\mathbf{y} = (z, \xi)$, $d\mathbf{y} = dz d\xi$.

ii. $f(z, \xi) = \frac{1}{(2\pi\varepsilon)^n} a(z)$.

iii. $\phi(\mathbf{y}) = \phi_\pm(x, z, \xi, t)$.

iv. The critical point is $\mathbf{y}_0 = (z_0, \xi_0)$.

3.3 **Substitute and Simplify:** Plugging our specific functions into the general formula:

$$I_\pm^\varepsilon(x,t) = \int f(\mathbf{y}) e^{\frac{i}{\varepsilon}\phi(\mathbf{y})} d\mathbf{y} \sim \frac{(2\pi\varepsilon)^n}{|\det H|^{1/2}} e^{\frac{i}{\varepsilon}\phi_\pm(\mathbf{y}_0) + i\frac{\pi}{4}\mathrm{sgn}(H)} \left(\frac{1}{(2\pi\varepsilon)^n} a(z_0) + O(\varepsilon) \right)$$
$$= \frac{a(z_0)}{|\det H|^{1/2}} e^{\frac{i}{\varepsilon}\phi_\pm(x, z_0, \xi_0, t) + i\frac{\pi}{4}\mathrm{sgn}(H)} + O(\varepsilon). \tag{2.139}$$

Note the crucial cancellation of the factor $(2\pi\varepsilon)^n$. This yields:

$$I_\pm^\varepsilon(x,t) \sim \frac{a(z_0)}{|\det H|^{1/2}} e^{\frac{i}{\varepsilon}\phi_\pm(x, z_0, \xi_0, t) + i\frac{\pi}{4}\mathrm{sgn}(H)} + O(\varepsilon). \tag{2.140}$$

4. *Step 4: Simplification of the Determinant:* The final step is to simplify the determinant of the Hessian, $\det(H)$, which appears in the amplitude term.

4.1 **Block Matrix Determinant Formula:** The Hessian H is a block matrix. A standard identity for the determinant of a block matrix is:

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(A) \det\Big(D - CA^{-1}B\Big). \tag{2.141}$$

provided A is invertible. Applying this to our Hessian (2.137) with $A = \nabla^2 p(z_0)$, $B = C = -I$, and $D = \pm t \left(\frac{I}{|\xi_0|} - \frac{\xi_0 \xi_0^T}{|\xi_0|^3} \right) = \pm \frac{t}{|\xi_0|} P(\xi_0)$:

$$\det(H) = \det\big(\nabla^2 p(z_0)\big) \cdot \det\left(\pm \frac{t}{|\xi_0|} P(\xi_0) - (-I)(\nabla^2 p(z_0))^{-1}(-I) \right)$$
$$= \det\big(\nabla^2 p(z_0)\big) \cdot \det\left(\pm \frac{t}{|\xi_0|} P(\xi_0) - (\nabla^2 p(z_0))^{-1} \right). \tag{2.142}$$

This form, while valid, is not the most geometrically insightful.

4.2 **From Block Matrix to Geometric Form:** We now manipulate Equation (2.142) to obtain Evans's result. The key is to factor the matrix $(\nabla^2 p(z_0))^{-1}$ out of the determinant.

4.2.1 **Factor out $-(\nabla^2 p(z_0))^{-1}$:** Note that:

$$\pm \frac{t}{|\xi_0|} P(\xi_0) - (\nabla^2 p(z_0))^{-1} = -(\nabla^2 p(z_0))^{-1} \left[I - \nabla^2 p(z_0) \left(\pm \frac{t}{|\xi_0|} P(\xi_0) \right) \right]. \tag{2.143}$$

For the forward wave case (+), this becomes:

$$\frac{t}{|\xi_0|} P(\xi_0) - (\nabla^2 p(z_0))^{-1} = -(\nabla^2 p(z_0))^{-1} \left[I - \frac{t}{|\xi_0|} \nabla^2 p(z_0) P(\xi_0) \right]. \tag{2.144}$$

4.2.2 **Apply the determinant identity:** Substituting back into (2.142):

$$\begin{aligned}\det(H) &= \det\big(\nabla^2 p(z_0)\big) \cdot \det\left(-(\nabla^2 p(z_0))^{-1}\left[I - \frac{t}{|\xi_0|}\nabla^2 p(z_0) P(\xi_0)\right]\right) \\ &= \det\big(\nabla^2 p(z_0)\big) \cdot \det\Big(-(\nabla^2 p(z_0))^{-1}\Big) \cdot \det\left(I - \frac{t}{|\xi_0|}\nabla^2 p(z_0) P(\xi_0)\right).\end{aligned} \tag{2.145}$$

4.2.3 **Simplify the constants:** Recall that for any invertible matrix M, $\det(-M) = (-1)^n \det(M)$. Therefore:

$$\det\Big(-(\nabla^2 p(z_0))^{-1}\Big) = (-1)^n \det\Big((\nabla^2 p(z_0))^{-1}\Big) = \frac{(-1)^n}{\det(\nabla^2 p(z_0))}. \tag{2.146}$$

Substituting this in:

$$\begin{aligned}\det(H) &= \det\big(\nabla^2 p(z_0)\big) \cdot \left(\frac{(-1)^n}{\det(\nabla^2 p(z_0))}\right) \cdot \det\left(I - \frac{t}{|\xi_0|}\nabla^2 p(z_0) P(\xi_0)\right) \\ &= (-1)^n \det\left(I - \frac{t}{|\xi_0|}\nabla^2 p(z_0) P(\xi_0)\right).\end{aligned} \tag{2.147}$$

4.2.4 **Final substitution:** On the stationary set, $\xi_0 = \nabla p(z_0)$, so $|\xi_0| = |\nabla p(z_0)|$. This yields the final result:

$$\det(H) = (-1)^n \det\left(I - \frac{t}{|\nabla p(z_0)|}\nabla^2 p(z_0) P(\nabla p(z_0))\right). \tag{2.148}$$

4.3 **Interpretation via Principal Curvatures:** The determinant in (2.148) contains the matrix expression that we can define as:

$$M := \frac{1}{|\nabla p(z_0)|}\nabla^2 p(z_0) P(\nabla p(z_0)). \tag{2.149}$$

The determinant then becomes:

$$\det(H) = (-1)^n \det(I - tM). \tag{2.150}$$

We now analyze the eigenvalue structure of M:

4.3.1 **Eigenvalue structure:** Since $P(\nabla p)$ is a projection matrix ($P^2 = P$) with $P(\nabla p)\nabla p = 0$, the matrix M has:

i. One zero eigenvalue (corresponding to eigenvector ∇p)

ii. $n-1$ nonzero eigenvalues $\lambda_1, \ldots, \lambda_{n-1}$

4.3.2 **Geometric meaning:** These nonzero eigenvalues are exactly the principal curvatures $\kappa_1, \ldots, \kappa_{n-1}$ of the level surface $\{p = p(z_0)\}$ at z_0. This follows because:

i. The product $\nabla^2 p(z_0) P(\nabla p(z_0))$ represents the restriction of the Hessian to the

tangent plane

ii. The scaling factor $1/|\nabla p(z_0)|$ converts this to the shape operator of the level surface

iii. The eigenvalues of the shape operator are by definition the principal curvatures

4.3.3 **Determinant calculation:** Since the determinant equals the product of eigenvalues:

$$\det(I - tM) = (1 - t \cdot 0) \prod_{i=1}^{n-1} (1 - t\lambda_i) = \prod_{i=1}^{n-1} (1 - t\kappa_i(z_0)) \tag{2.151}$$

Substituting into (2.148) gives the final result:

$$\det(H) = (-1)^n \prod_{i=1}^{n-1} (1 - t\kappa_i(z_0)) \tag{2.152}$$

This result reveals the geometric optics behavior: the amplitude scales as $\prod_{i=1}^{n-1} |1 - t\kappa_i|^{-1/2}$, showing focusing ($\kappa_i > 0$) or defocusing ($\kappa_i < 0$) of rays. Caustics form when $1 - t\kappa_i = 0$, where the asymptotic description breaks down.

5. *Step 5: Final Result and Physical Interpretation.* We now combine the results from the previous steps to write the final asymptotic formula and interpret it.

5.1 **Final Asymptotic Formula:** Substituting the determinant (2.152) back into the stationary phase result (2.140), we obtain:

$$\begin{aligned} I_{\pm}^{\varepsilon}(x,t) &\sim \frac{a(z_0)}{\left|(-1)^n \prod_{i=1}^{n-1}(1 - t\kappa_i(z_0))\right|^{1/2}} e^{\frac{i}{\varepsilon}\phi_{\pm}(x,z_0,\xi_0,t) + i\frac{\pi}{4}\operatorname{sgn}(H)} + O(\varepsilon) \\ &= \frac{a(z_0)}{\sqrt{\prod_{i=1}^{n-1} |1 - t\kappa_i(z_0)|}} e^{\frac{i}{\varepsilon}\phi_{\pm}(x,z_0,\xi_0,t) + i\frac{\pi}{4}(\operatorname{sgn}(H)+n)} + O(\varepsilon). \end{aligned} \tag{2.153}$$

The phase shift simplifies to a constant $\frac{\pi}{4}(\operatorname{sgn}(H) + n)$, which can be written as $\frac{\pi}{4}\sigma$ for some integer σ. The expression becomes:

$$I_{\pm}^{\varepsilon}(x_0,t_0) = \frac{e^{\frac{i\pi}{4}\sigma}}{|\det D_{y,z}^2 \phi_{\pm}|^{1/2}} e^{\frac{i}{\varepsilon}\phi_{\pm}} \left(a(z_0) + O(\varepsilon)\right), \tag{2.154}$$

which matches the expression in Evans. The full solution $u^{\varepsilon}(x_0,t_0)$ is given by the average of the contributions from I_+^{ε} and I_-^{ε}.

2.5 Proofs of Supporting Lemmas (Appendices)

LEMMA 3 (Changing coordinates). *Assume* $\phi : \mathbb{R}^n \to \mathbb{R}$ *is smooth.*

(i) **(Part i)** *Suppose that* $D\phi(0) \neq 0$. *Then there exists a smooth function* $\mathbf{\Phi} : \mathbb{R}^n \to \mathbb{R}^n$ *such that*

$$\begin{cases} \mathbf{\Phi}(0) = 0, \quad D\mathbf{\Phi}(0) = I, \\ \phi(\mathbf{\Phi}(x)) = \phi(0) + D\phi(0) \cdot x \quad \text{for } |x| \text{ small.} \end{cases}$$

This means we can change variables near 0 to make ϕ affine (linear + constant).

(ii) **(Part ii, Morse Lemma)** *Suppose instead that* $D\phi(0) = 0$, $D^2\phi(0) \neq 0$. *Then there exists a smooth function* $\mathbf{\Phi} : \mathbb{R}^n \to \mathbb{R}^n$ *such that*

$$\begin{cases} \mathbf{\Phi}(0) = 0, \quad D\mathbf{\Phi}(0) = I, \\ \phi(\mathbf{\Phi}(x)) = \phi(0) + \frac{1}{2} x \cdot D^2\phi(0) x \quad \text{for } |x| \text{ small.} \end{cases}$$

This means we can change variables near 0 to make ϕ purely quadratic [a]

[a] The analysis is centered at 0 for simplicity. For an arbitrary point y_0, we can shift $\phi(y)$ to $\tilde{\phi}(y) = \phi(y + y_0)$; all derivatives at 0 then match those of ϕ at y_0.

Geometric intuition: If we think of $\mathbf{\Phi}$ as "reshaping" the level sets of ϕ near 0: for part (i), it flattens curved level sets into hyperplanes, and for part (ii), it transforms curved level sets into quadratic surfaces.

2.5.1 *Proof of Lemma 3(i).*

Proof.

1. *Step 1: Constructing the System:* Given $D\phi(0) = r_n \neq 0$, we complete it to an orthogonal basis $\{r_1, \ldots, r_n\}$ of $\mathbb{R}^n$. The purpose is to isolate the direction of $D\phi(0)$ from its orthogonal complement. Define the map $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ by:

$$f(x, y) = \begin{pmatrix} r_1 \cdot (y - x) \\ \vdots \\ r_{n-1} \cdot (y - x) \\ \phi(y) - \phi(0) - D\phi(0) \cdot x \end{pmatrix}. \tag{2.155}$$

where $x \in \mathbb{R}^n$ (the input space) is the original coordinates, and $y \in \mathbb{R}^n$ (the output space to solve for) is the new coordinates. The total domain is $\mathbb{R}^n \times \mathbb{R}^n$, so f can compute derivatives with respect to both x and y.

   i. First $n - 1$ components: enforce $y - x \perp \text{span}\{r_1, \cdots, r_{n-1}\}$ (e.g., $y - x \parallel D\phi(0)$).

ii. The last component n: measures deviation from linear approximation.

2. *Step 2: Apply the Implicit Function Theorem:* The IFT is applied to solve $f(x, y) = f(x, \mathbf{\Phi}(x)) = 0$, where $y = \mathbf{\Phi}(x)$.

2.1 **Recall the IFT Statement:**

> **Implicit Function Theorem (IFT) Statement.** *Given $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ of class C^1 with $f(0,0) = 0$, if $D_y f(0,0)$ is invertible, then:*
>
> - *There exists a neighborhood $U \subset \mathbb{R}^n$ of 0 and a C^1 map $\mathbf{\Phi} : U \to \mathbb{R}^n$:*
>
> (a) $\mathbf{\Phi}(0) = 0$,
>
> (b) $f(x, \mathbf{\Phi}(x)) = 0$ for all $x \in U$,
>
> (c) $D\mathbf{\Phi}(0) = -[D_y f(0,0)]^{-1} D_x f(0,0)$.
>
> **Note:** the complete version of IFT can be found at the Sec (**??**).

2.2 **Verify IFT Conditions for Our Case:**

i. **Smoothness:** f is C^1 because ϕ is smooth and dot products are linear.

ii. **Initial condition:** We can get $f(0,0) = 0$ by direct computation:

$$f(0,0) = (r_1 \cdot 0, \cdots, r_{n-1} \cdot 0, \phi(0) - \phi(0) - 0) = 0. \tag{2.156}$$

iii. **Invertibility of $D_y f(0,0)$:** We can compute it as the follow:

$$D_y f(0,0) = \begin{pmatrix} \nabla_y (r_1 \cdot (y - x))\big|_{(0,0)} \\ \vdots \\ \nabla_y (r_{n-1} \cdot (y - x))\big|_{(0,0)} \\ \nabla_y (\phi(y) - \phi(0) - D\phi(0) \cdot x)\big|_{(0,0)} \end{pmatrix} = \begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix}. \tag{2.157}$$

where the last row follows because $\nabla_y \phi(y)\big|_{y=0} = D\phi(0) = r_n$. Moreover, since $\{r_k\}_{k=1}^n$ is an orthogonal basis, the rows are linearly independent, and $\det\big(D_y f(0,0)\big) \neq 0$. $D_y f(0,0)$ is then invertible.

2.3 **Conclusion:** The IFT guarantees the existence of $\mathbf{\Phi}$ with:

i. **Initial condition:** $\mathbf{\Phi}(0) = 0$ ($f(0,0) = 0$ and $\mathbf{\Phi}(0)$ must satisfy $f(0, \mathbf{\Phi}(0)) = 0$).

ii. **Solution property:** $f(x, \mathbf{\Phi}(x)) = 0$ for $x \in U$, which expands to:

$$\begin{cases} r_k \cdot (\mathbf{\Phi}(x) - x) = 0 & \text{(orthogonality constraints)}, \\ \phi(\mathbf{\Phi}(x)) = \phi(0) + D\phi(0) \cdot x & \text{(affine condition)}. \end{cases} \tag{2.158}$$

iii. **Derivative formula:**

$$D\mathbf{\Phi}(0) = -\left[D_y f(0,0)\right]^{-1} D_x f(0,0). \tag{2.159}$$

Overall, the theorem guarantees such a solution $\mathbf{\Phi}$ exists locally and is smooth.

3. *Step 3: Analyze Transformed Coordinates:* From $f(x, \mathbf{\Phi}(x)) = 0$, we can extract two critical conditions:

   3.1 **Orthogonality Constraints:**

   $$r_k \cdot (\mathbf{\Phi}(x) - x) = 0 \quad \text{for } k = 1, \ldots, n-1. \tag{2.160}$$

   Geometrically, the difference $\mathbf{\Phi}(x) - x$ must be orthogonal to all basis vectors except $r_n = D\phi(0)$. Thus, it is parallel to $D\phi(0)$:

   $$\mathbf{\Phi}(x) - x = \alpha(x) D\phi(0) \quad \text{for some scalar function } \alpha(x). \tag{2.161}$$

   This forces the transformation $\mathbf{\Phi}$ to only "shift" points along the $D\phi(0)$ direction.

   3.2 **Affine Condition:**

   $$\phi(\mathbf{\Phi}(x)) = \phi(0) + D\phi(0) \cdot x. \tag{2.162}$$

   ϕ becomes *exactly linear* in the new coordinates. This condition enforces that ϕ at $\mathbf{\Phi}(x)$ matches the linear approximation at x. With the orthogonality constraints, it ensure $\mathbf{\Phi}$ compensates for the nonlinearity of ϕ.

4. *Step 4: Verify $D\mathbf{\Phi}(x)$:* To confirm $\mathbf{\Phi}$ is well-behaved at the origin:

   4.1 **Differentiate $f(x, \mathbf{\Phi}(x)) = 0$ at $x = 0$:**

   $$\underbrace{D_x f(0,0)}_{\text{Jacobian w.r.t. } x} + \underbrace{D_y f(0,0)}_{\text{Jacobian w.r.t. } y} \cdot D\mathbf{\Phi}(0) = 0. \tag{2.163}$$

   4.2 **Compute $D_x f(0,0)$:**

   $$D_x f(0,0) = \begin{pmatrix} \nabla_x \left(r_1 \cdot (y - x)\right) \\ \vdots \\ \nabla_x \left(r_{n-1} \cdot (y - x)\right) \\ \nabla_x \left(\phi(y) - \phi(0) - D\phi(0) \cdot x\right) \end{pmatrix}_{(0,0)} = \begin{pmatrix} -r_1 \\ \vdots \\ -r_{n-1} \\ -D\phi(0) \end{pmatrix} = -D_y f(0,0). \tag{2.164}$$

   The last equality holds because $D_y f(0,0) = \begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix}$.

4.3 **Substitute and solve:**

$$-D_y f(0,0) + D_y f(0,0) D\mathbf{\Phi}(0) = 0 \implies D\mathbf{\Phi}(0) = I. \tag{2.165}$$

So, overall we have $\mathbf{\Phi}(0) = 0$ (origin preserved), and $D\mathbf{\Phi}(0) = I$ (no distortion at first order). Thus, ϕ becomes perfectly linear in new coordinates. □

2.5.2 Proof of Lemma 3(ii) (Morse Lemma).

Proof.

1. *Step 1: Quadratic Expansion*:

   Given $\phi \in C^2(\mathbb{R}^n)$ with $D\phi(0) = 0$, and $D^2\phi(0)$ nonsingular, we want to express $\phi(y)$ as:

$$\phi(y) = \phi(0) + \frac{1}{2} y \cdot A(y) y. \tag{2.166}$$

   where $A(y)$ is smooth, symmetric, and $A(0) = D^2\phi(0)$.

   1.1 **Single-Variable Case:** For intuition, we first consider $\phi : \mathbb{R} \to \mathbb{R}$ with $\phi'(0) = 0$.

   1.1.1 **Fundamental Theorem of Calculus:** For any C^1 function ϕ, we can write:

$$\phi(y) = \phi(0) + \int_0^y \phi'(t) dt. \tag{2.167}$$

   Since $\phi'(0) = 0$, the linear term vanishes.

   1.1.2 **Integration by Parts:** To expose the second derivative for C^2, we apply integration by parts:

$$\int_0^y \phi'(t) dt = \underbrace{[\phi'(t)\,(t-y)]_0^y}_{\substack{t=0:\ \phi'(0)=0 \\ \text{(critical point condition)}}} - \int_0^y (t-y)\phi''(t) dt. \tag{2.168}$$

   Thus, we have:

$$\phi(y) = \phi(0) + \int_0^y (y-t)\phi''(y) dt. \tag{2.169}$$

   For $y \geq 0$, this is equivalent to:

$$\phi(y) = \phi(0) + \int_0^1 (y - ty)\phi''(ty) \cdot y dt = \phi(0) + \frac{1}{2} y^2 \cdot \underbrace{2 \int_0^1 (1-t)\phi''(ty) dt}_{A(y)}. \tag{2.170}$$

   Then, we have:

$$\phi(y) = \phi(0) + \frac{1}{2} A(y) y^2. \tag{2.171}$$

   where $A(0) = \phi''(0)$, and $A(y)$ is smooth if $\phi \in C^2$.

1.2 **Multivariable Generalization:** For $\phi : \mathbb{R}^n \to \mathbb{R}$, fix a direction $y \in \mathbb{R}^n$ and define $g(t) = \phi(ty)$ for $t \in [0, 1]$.

1.2.1 **Apply Single-Variable Result to $g(t)$:**

$$g(1) = g(0) + \int_0^1 (1-t)g''(t)dt. \tag{2.172}$$

Here, $g(1) = \phi(y)$, $g(0) = \phi(0)$, and we compute derivatives of g:

$$g'(t) = D\phi(ty) \cdot y, \quad g''(t) = y \cdot D^2\phi(ty)y. \tag{2.173}$$

Since $D\phi(0) = 0$, we have $g'(0) = 0$, so:

$$\phi(y) = \phi(0) + \int_0^1 (1-t)y \cdot D^2\phi(ty)ydt. \tag{2.174}$$

1.2.2 **Rewrite as a Quadratic Form:** We can express $\phi(y)$ as:

$$\phi(y) = \phi(0) + \frac{1}{2}y \cdot \underbrace{\left(2\int_0^1 (1-t)D^2\phi(ty)dt\right)}_{A(y)} y. \tag{2.175}$$

Then, we have:

$$\phi(y) = \phi(0) + \frac{1}{2}y \cdot A(y)y. \tag{2.176}$$

where $A(y)$ is symmetric because $D^2\phi$ is symmetric. $A(0) = D^2\phi(0)$, and $A(y)$ is smooth in y because $\phi \in C^2$.

1.2.3 **Diagonalization of $A(0)$:** By assumption, $D^2\phi(0)$ is nonsingular, so its eigenvalues $\lambda_1, \cdots, \lambda_n$ are nonzero. We can then rotate coordinates to diagonalize $A(0)$:

$$A(0) = D^2\phi(0) = \text{diag}(\lambda_1, \cdots, \lambda_n), \quad \lambda_i \neq 0. \tag{2.177}$$

This rotation simplifies the inductive diagonalization in next steps.

2. *Step 2: Inductive Goal (Morse Lemma Proof):* We now proceed to the inductive construction of the coordinate transformation $\mathbf{\Phi}$. The goal is to iteratively eliminate cross-terms in the quadratic expansion of $\mathbf{\Phi}$ until only diagonal terms remain.

2.1 **Inductive Hypothesis:** For each $m \in \{0, \ldots, n\}$, we aim to construct a smooth coordinate change Φ_m such that:

$$\phi(\Phi_m(x)) = \underbrace{\phi(0)}_{\text{Term 1}} + \underbrace{\frac{1}{2}\sum_{i=1}^{m} \lambda_i x_i^2}_{\text{Term 2}} + \underbrace{\frac{1}{2}\sum_{i,j=m+1}^{n} a_m^{ij}(x)x_i x_j}_{\text{Term 3}}. \tag{2.178}$$

where λ_i are the eigenvalues of $D^2\phi(0)$ (from *Step 1*), $A_m(x) = (a_m^{ij}(x))$ is symmetric matrix with smooth coefficiens, and $a_m^{ij}(0)$ is diagonal: $a_m^{ij} = \lambda_i \delta_{ij}$ for $i, j \geq m+1$.

Key idea: The Morse Lemma constructs a coordinate change $\mathbf{\Phi}$ to diagonalize the quadratic part of ϕ one direction at a time.

At step m, we assume:

i. The first m directions $x_1, \cdots, x_m$ are already diagonalized (Term 2).

ii. The remaining directions $(x_{m+1}, \cdots, x_n)$ still have cross-terms (Term 3).

2.2 **Base Case** $(m = 0)$**:** Let $\Phi_0(x) = x$ (identity map). From *Step 1*, we have:

$$\phi(\Phi_0(x)) = \phi(0) + \frac{1}{2} x \cdot A_0(x) x, \quad \text{where } A_0(x) = A(x). \tag{2.179}$$

This satisfies the inductive hypothesis with $A_0(0) = D^2\phi(0) = \operatorname{diag}(\lambda_1, \cdot, \lambda_n)$. Thus, the inductive hypothesis holds trivially for $m = 0$.

2.3 **Inductive Step (**$m \to m+1$**):** Assume the hypothesis holds for some $m < n$. We now construct $\mathbf{\Phi}_{m+1}$ to eliminate the $(m+1)$-th cross-terms.

2.3.1 **Coordinate Change:** We define a new coordinate system $\Pi_{m+1}(y)$ as:

$$\Pi_{m+1}(y) = \left(y_1, \ldots, y_m, \underbrace{\sqrt{\frac{a_m^{m+1,m+1}(y)}{\lambda_{m+1}}}}_{①} \left(y_{m+1} + \underbrace{\sum_{j=m+2}^{n} \frac{y_j a_m^{m+1,j}(y)}{a_m^{m+1,m+1}(y)}}_{②} \right), y_{m+2}, \ldots, y_n \right). \tag{2.180}$$

where ① scales the $(m+1)$-th coordinate to normalize its coefficient to λ_{m+1}, and the sum ② eliminates cross-terms involving y_{m+1}.

2.3.2 **Key Properties of** $\prod_{m+1}(y)$**:**

i. **Smoothness:** Since $a_m^{m+1,m+1}(0) = \lambda_{m+1} \neq 0$, the denominator is nonzero near $y = 0$.

ii. **Preservation of Origin:** $\prod_{m+1}(0) = 0$.

iii. **Jacobian at** 0**:** $D\prod_{m+1}(0) = I$ (identity matrix), ensuring the transformation is invertible near 0.

2.3.3 **Inductive Transformation:** Let $\Xi_{m+1} = \Pi_{m+1}^{-1}$ (the inverse exists locally by the Inverse Function Theorem) and define

$$\Phi_{m+1} = \Phi_m \circ \Xi_{m+1}. \tag{2.181}$$

Applying Ξ_{m+1} to the inductive hypothesis for m yields:

$$\phi\big(\Phi_m(\Xi_{m+1}(x))\big) = \phi(0) + \frac{1}{2} \sum_{i=1}^{m} \lambda_i x_i^2 + \frac{1}{2} \sum_{i,j=m+1}^{n} a_m^{ij}\big(\Xi_{m+1}(x)\big)\, \xi_i\, \xi_j. \tag{2.182}$$

where $\xi = \Xi_{m+1}(x)$. The coordinate change Π_{m+1} is designed so that

$$\sum_{i,j=m+1}^{n} a_m^{ij}\big(\Xi_{m+1}(x)\big)\,\xi_i\,\xi_j = \lambda_{m+1}\,x_{m+1}^2 + \sum_{i,j=m+2}^{n} b_m^{ij}(x)\,x_i\,x_j. \tag{2.183}$$

where $b_m^{ij}(x)$ are new coefficients with $b_m^{ij}(0) = \lambda_i\,\delta_{ij}$.

2.3.4 **Update the Quadratic Form:** The transformed ϕ becomes:

$$\phi\big(\Phi_{m+1}(x)\big) = \phi(0) + \frac{1}{2}\sum_{i=1}^{m+1}\lambda_i x_i^2 + \frac{1}{2}\sum_{i,j=m+2}^{n} b_m^{ij}(x)x_i x_j. \tag{2.184}$$

The $(m+1)$-th diagonal term is now isolated as $\lambda_{m+1}\,x_{m+1}^2$, and the remaining cross-terms (for $i, j \geq m+2$) are handled in the next inductive step.

2.4 **Final Step** $(m = n)$**:** After n steps, all cross-terms are eliminated:

$$\phi\big(\Phi_n(x)\big) = \phi(0) + \frac{1}{2}\sum_{i=1}^{n}\lambda_i x_i^2 = \phi(0) + \frac{1}{2}\,x\cdot D^2\phi(0)x. \tag{2.185}$$

□

2.5.3 Statement of the Implicit Function Theorem

1. **Preliminaries:** Let n, m be positive integers. We write a typical point in $\mathbb{R}^{n+m}$ as

$$(x, y) = (x_1, \ldots, x_n, y_1, \ldots, y_m), \quad \text{where } x \in \mathbb{R}^n,\ y \in \mathbb{R}^m. \tag{2.186}$$

2. **Setup:**

   i. Let $U \subset \mathbb{R}^{n+m}$ be an open set

   ii. Let $f : U \to \mathbb{R}^m$ be a C^1 function, where $f = (f^1, \ldots, f^m)$

   iii. Let $(x_0, y_0) \in U$ be a point, and define $z_0 = f(x_0, y_0)$

3. **Jacobian Matrix:** The gradient matrix (Jacobian) of f is:

$$Df = \begin{pmatrix} \frac{\partial f^1}{\partial x_1} & \cdots & \frac{\partial f^1}{\partial x_n} & \frac{\partial f^1}{\partial y_1} & \cdots & \frac{\partial f^1}{\partial y_m} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x_1} & \cdots & \frac{\partial f^m}{\partial x_n} & \frac{\partial f^m}{\partial y_1} & \cdots & \frac{\partial f^m}{\partial y_m} \end{pmatrix}_{m\times(n+m)} = \big(D_x f \mid D_y f\big). \tag{2.187}$$

4. **Critical Condition:** The Jacobian determinant with respect to y is:

$$J_y f(x_0, y_0) = \left|\det D_y f(x_0, y_0)\right| = \left|\det \begin{pmatrix} \frac{\partial f^1}{\partial y_1} & \cdots & \frac{\partial f^1}{\partial y_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial y_1} & \cdots & \frac{\partial f^m}{\partial y_m} \end{pmatrix}\right|. \tag{2.188}$$

5. **Implicit Function Theorem:** Assume $f \in C^1(U; \mathbb{R}^m)$ and $J_y f(x_0, y_0) \neq 0$. Then:

   1. There exist open neighborhoods:
      (a) $V \subset U$ with $(x_0, y_0) \in V$
      (b) $W \subset \mathbb{R}^n$ with $x_0 \in W$
   2. There exists a C^1 mapping $g : W \to \mathbb{R}^m$ such that:
      (a) $g(x_0) = y_0$
      (b) $f(x, g(x)) = z_0$ for all $x \in W$
      (c) If $(x, y) \in V$ and $f(x, y) = z_0$, then $y = g(x)$
   3. (Regularity) If $f \in C^k(U; \mathbb{R}^m)$, then $g \in C^k(W; \mathbb{R}^m)$ for $k = 2, 3, \ldots$

   The function g is *implicitly defined* near x_0 by the equation $f(x, y) = z_0$.

3 HOMOGENIZATION

3.1 Periodic homogenization of an elliptic equation

Step 1: The System Overview

We consider the elliptic PDE with rapidly oscillating coefficients:

$$L^\varepsilon u^\varepsilon = -\sum_{i,j} \frac{\partial}{\partial x_j}\left(a^{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u^\varepsilon}{\partial x_i}\right) = f(x) \quad \text{in } U. \tag{3.1}$$

subject to the Dirichlet boundary condition $u^\varepsilon = 0$ on ∂U. The coefficients $a^{ij}(y)$ are Q-periodic (where $Q = [0,1]^n$ is the unit cube) and uniformly elliptic, meaning they satisfy $a^{ij}(y+k) = a^{ij}(y)$ for any integer vector k.

Here, $y = x/\varepsilon$ is the fast variable, and u^ε denotes the oscillatory solution induced by the fine-scale structure of the coefficients.

Step 2: Two-Scale Ansatz

Assume a two-scale expansion (ansatz) of the form

$$u^\varepsilon(x) = u_0(x,y) + \varepsilon\, u_1(x,y) + \varepsilon^2\, u_2(x,y) + \cdots, \quad \text{where } y = \frac{x}{\varepsilon}. \tag{3.2}$$

where each term $u_k(x,y)$ depends on the macroscopic variable x and the microscopic variable $y = x/\varepsilon$. For all $k \geq 0$, the functions $u_k(x,y)$ are assumed to be Q-periodic in y.

Note: This separation of scales models physical materials with periodic microstructures, such as composites or crystalline solids.

Step 3: Rewrite the Differential Operator

3.1. **Chain Rule for Multiscale Functions**

For a function $u(x,y)$ where $y = x/\varepsilon$, the partial derivative with respect to x_i becomes:

$$\frac{\partial}{\partial x_i}u(x,y) = \underbrace{\frac{\partial u}{\partial x_i}\bigg|_{y \text{ fixed}}}_{\text{Macroscopic change}} + \underbrace{\frac{\partial y_j}{\partial x_i}\frac{\partial u}{\partial y_j}}_{\text{Microscopic change}} \tag{3.3}$$

Since $y_j = x_j/\varepsilon$, we have $\frac{\partial y_j}{\partial x_i} = \frac{1}{\varepsilon}\delta_{ij}$, leading to:

$$\frac{\partial}{\partial x_i} \implies \frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}. \tag{3.4}$$

This is exact, not an approximation.

3.2. **Transforming the Original PDE**

Recall, we have (3.1), rewrite it in terms of $y = x/\varepsilon$ for the operator:

$$L^\varepsilon u^\varepsilon = -\sum_{i,j} \frac{\partial}{\partial x_j}\left(a^{ij}\left(\frac{x}{\varepsilon}\right)\frac{\partial u^\varepsilon}{\partial x_i}\right) = f(x) \implies L^\varepsilon = -\sum_{i,j}\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial_{x_i}}\right) = f(x). \tag{3.5}$$

3.3. **Substitute the Chain Rule into L^ε**

Replace each $\frac{\partial}{\partial x_k}$ with $\left(\frac{\partial}{\partial x_k} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_k}\right)$.

i. Inner Derivative:

$$\frac{\partial u^\varepsilon}{\partial x_i} \implies \left(\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\right)u^\varepsilon. \tag{3.6}$$

ii. Outer Derivative:

$$\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial u^\varepsilon}{\partial x_i}\right) \implies \left(\frac{\partial}{\partial x_j} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_j}\right)\left(a^{ij}(y)\left(\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\right)\right)u^\varepsilon. \tag{3.7}$$

Hence, we have the two-scale differential operator as:

$$L^\varepsilon = -\sum_{i,j}\left(\frac{\partial}{\partial x_j} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_j}\right)\left(a^{ij}(y)\left(\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\right)\right). \tag{3.8}$$

3.4. **Expansion of L^ε in powers of ε**

3.4.1 **Expand the operator:** First expand the inner parentheses:

$$a^{ij}(y)\left(\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_i}\right) = a^{ij}(y)\frac{\partial}{\partial x_i} + \frac{a^{ij}(y)}{\varepsilon}\frac{\partial}{\partial y_i}. \tag{3.9}$$

Now, apply the outer derivative:

$$L^\varepsilon = -\sum_{i,j}\left(\frac{\partial}{\partial x_j} + \frac{1}{\varepsilon}\frac{\partial}{\partial y_j}\right)\left(a^{ij}(y)\frac{\partial}{\partial x_i} + \frac{a^{ij}(y)}{\varepsilon}\frac{\partial}{\partial y_i}\right). \tag{3.10}$$

3.4.2 **Distribute term-by-term:** Distribute the outer derivative:

$$L^\varepsilon = -\sum_{i,j}\Bigg[\underbrace{\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right)}_{\text{Term A}} + \underbrace{\frac{1}{\varepsilon}\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial y_i}\right)}_{\text{Term B}} + \underbrace{\frac{1}{\varepsilon}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right)}_{\text{Term C}} + \underbrace{\frac{1}{\varepsilon^2}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial y_i}\right)}_{\text{Term D}}\Bigg]. \tag{3.11}$$

3.4.3. **Simplify each term:**

a. **Term A:**

$$\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right) = \frac{\partial a^{ij}(y)}{\partial x_j}\frac{\partial}{\partial x_i} + a^{ij}(y)\frac{\partial^2}{\partial x_j \partial x_i}. \tag{3.12}$$

Since $a^{ij}(y)$ depends on x only through $y = x/\varepsilon$, $\frac{\partial a^{ij}(y)}{\partial x_j} = \frac{1}{\varepsilon}\frac{\partial a^{ij}(y)}{\partial y_j}$. Hence:

$$\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right) = \frac{1}{\varepsilon}\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial x_i} + a^{ij}(y)\frac{\partial^2}{\partial x_j \partial x_i}. \tag{3.13}$$

b. **Term B:**

$$\begin{aligned}\frac{1}{\varepsilon}\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial y_i}\right) &= \frac{1}{\varepsilon}\left[\frac{\partial a^{ij}(y)}{\partial x_j}\frac{\partial}{\partial y_i} + a^{ij}(y)\frac{\partial^2}{\partial x_j \partial y_i}\right] \\ &= \frac{1}{\varepsilon^2}\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial y_i} + \frac{1}{\varepsilon}a^{ij}(y)\frac{\partial^2}{\partial x_j \partial y_i}.\end{aligned} \tag{3.14}$$

c. **Term C:**

$$\frac{1}{\varepsilon}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right) = \frac{1}{\varepsilon}\left[\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial x_i} + a^{ij}(y)\frac{\partial^2}{\partial y_j \partial x_i}\right]. \tag{3.15}$$

d. **Term D:**

$$\frac{1}{\varepsilon^2}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial y_i}\right) = \frac{1}{\varepsilon^2}\left[\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial y_i} + a^{ij}(y)\frac{\partial^2}{\partial y_j \partial y_i}\right]. \tag{3.16}$$

3.4.4 Combine and group by powers of ε:

① : $\mathcal{O}(\varepsilon^{-2})$ terms:

$$\frac{1}{\varepsilon^2}\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial y_i} + \frac{1}{\varepsilon^2}a^{ij}(y)\frac{\partial^2}{\partial y_j \partial y_i} = \frac{1}{\varepsilon^2}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial y_i}\right). \tag{3.17}$$

This defines:

$$L_1 = -\sum_{i,j}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial y_i}\right). \tag{3.18}$$

② : $\mathcal{O}(\varepsilon^{-1})$ terms:

$$\frac{1}{\varepsilon}\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial x_i} + \frac{1}{\varepsilon}a^{ij}(y)\frac{\partial^2}{\partial x_j \partial y_i} + \frac{1}{\varepsilon}a^{ij}(y)\frac{\partial^2}{\partial y_j \partial x_i}. \tag{3.19}$$

Noting $\frac{\partial^2}{\partial x_j \partial y_i} = \frac{\partial^2}{\partial y_i \partial x_j}$, this combines to

$$\frac{2}{\varepsilon}\frac{\partial a^{ij}(y)}{\partial y_j}\frac{\partial}{\partial x_i} + \frac{2}{\varepsilon}a^{ij}(y)\frac{\partial^2}{\partial y_j \partial x_i} \implies \frac{2}{\varepsilon}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right). \tag{3.20}$$

This defines

$$L_2 = -\sum_{i,j}\left[\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right) + \frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial y_i}\right)\right]. \tag{3.21}$$

③ : $\mathcal{O}(1)$ terms:

$$a^{ij}(y)\frac{\partial^2}{\partial x_j \partial x_i}. \tag{3.22}$$

This defines

$$L_3 = -\sum_{i,j}\frac{\partial}{\partial x_j}\left(a^{ij}(y)\frac{\partial}{\partial x_i}\right). \tag{3.23}$$

Step 4: Solve the $\mathcal{O}(\varepsilon^{-2})$ Equation

4.1 **Dominant Balance at $\mathcal{O}(\varepsilon^{-2})$**

For the equation to hold as $\varepsilon \to 0$, the coefficient of each power of ε must vanish. The most singular term is $\mathcal{O}(\varepsilon^{-2})$, so we set

$$L_1 u_0 = 0. \tag{3.24}$$

This is a necessary condition to avoid blow-up as $\varepsilon \to 0$.

4.2 **Solving $L_1 u_0 = 0$**

Recall what we have for the leading-order equation

$$L_1 u_0 \;=\; -\sum_{i,j}\frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial u_0}{\partial y_i}\right) \;=\; 0. \tag{3.25}$$

This is a second-order elliptic PDE in y for fixed x, with unknown: $u_0(x, y)$, and boundary conditions: u_0 is Q-periodic in y, where $Q = [0, 1]^n$ is the unit cell.

4.2.1 **Rewrite the Equation for Clarity:** Define the flux term, then (3.1) becomes:

$$F^i(y) = a^{ij}(y)\frac{\partial u_0}{\partial y_i} \implies L_1 u_0 = -\sum_j \frac{\partial}{\partial y_j}F^j(y) = 0. \tag{3.26}$$

4.2.2 **Multiply by a Test Function and Integrate:** Multiply by the test function u_0 and integrate over the unit cell $Q = [0, 1]^n$:

$$-\int_Q u_0 \sum_j \frac{\partial F^j}{\partial y_j}\, dy = 0. \tag{3.27}$$

4.2.3 **Apply Integration by Parts:** For each j, apply IBP to the term $\int_Q u_0 \frac{\partial F^j}{\partial y_j} dy$. We then identify u and v as:

① $u = u_0$ (the test function); ② $v_j = F^j(y) = a^{ij}(y)\,\partial_{y_i} u_0$ (the flux).

In 1D IBP analogy:

$$\int u\,\frac{dv}{dy}\,dy = uv\Big|_{\text{boundary}} - \int \frac{du}{dy}\,v\,dy. \tag{3.28}$$

In n dimensions, for each j, the divergence theorem gives [15]:

$$\int_Q u_0 \frac{\partial v_j}{\partial y_j}\, dy = \underbrace{\int_{\partial Q} u_0\, v_j\, n_j\, dS}_{\text{Boundary term}} - \int_Q \frac{\partial u_0}{\partial y_j}\, v_j\, dy. \tag{3.29}$$

where n_j is the outward unit normal on ∂Q. By Q-periodicity of u_0 and $a^{ij}(y)$, the boundary terms cancel. Hence for each j

$$\int_Q u_0 \frac{\partial F^j}{\partial y_j}\, dy = -\int_Q \frac{\partial u_0}{\partial y_j} F^j(y)\, dy \implies \int_Q u_0 \frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial u_0}{\partial y_i}\right) dy = -\int_Q \frac{\partial u_0}{\partial y_j}\, a^{ij}(y)\, \frac{\partial u_0}{\partial y_i}\, dy. \tag{3.30}$$

with $F^j = a^{ij}(y)\, \partial_{y_i} u_0$ substituted back.

4.2.4 **Sum Over j and Substitute Back:** Sum over j and negate (from the original equation)

$$-\sum_j \int_Q u_0 \frac{\partial}{\partial y_j}\left(a^{ij}(y)\frac{\partial u_0}{\partial y_i}\right) dy = \sum_j \int_Q \frac{\partial u_0}{\partial y_j}\, a^{ij}(y)\, \frac{\partial u_0}{\partial y_i}\, dy. \tag{3.31}$$

But the left-hand side is 0 from *step 4.2.2*. Therefore

$$\sum_{i,j} \int_Q \frac{\partial u_0}{\partial y_j}\, a^{ij}(y)\, \frac{\partial u_0}{\partial y_i}\, dy = 0. \tag{3.32}$$

4.3.5 **Use uniform ellipticity:** By assumption, $a^{ij}(y)$ is uniformly elliptic on Q: there exists $\theta > 0$ such that for all $\xi \in \mathbb{R}^n$ and all $y \in Q$ [16]:

$$\sum_{i,j} a^{ij}(y)\, \xi_i \xi_j \;\geq\; \theta \|\xi\|^2, \quad \text{for some } \theta > 0. \tag{3.33}$$

where $\|\xi\|^2 = \xi_1^2 + \cdots + \xi_n^2$. Take $\xi = \nabla_y u_0(y)$ pointwise. Then:

$$\sum_{i,j} \frac{\partial u_0}{\partial y_j}\, a^{ij}(y) \frac{\partial u_0}{\partial y_i} \;\geq\; \theta \left\|\nabla_y u_0\right\|^2. \tag{3.34}$$

Integrate over Q:

$$\int_Q \sum_{i,j} \frac{\partial u_0}{\partial y_j}\, a^{ij}(y) \frac{\partial u_0}{\partial y_i}\, dy \;\geq\; \theta \int_Q \|\nabla_y u_0\|^2\, dy. \tag{3.35}$$

[15] The divergence theorem in $\mathbb{R}^n$ states that for any sufficiently smooth vector field $\mathbf{v}$ and scalar field u_0 on a domain Q with boundary ∂Q: $\int_Q u_0(\nabla \cdot \mathbf{v})\, dy = \int_{\partial Q} u_0(\mathbf{v} \cdot \mathbf{n})\, dS - \int_Q \nabla u_0 \cdot \mathbf{v}\, dy$. For Q-periodic functions, the boundary term vanishes because contributions from opposite faces cancel exactly due to periodicity.

[16] The uniform ellipticity condition ensures that energy dissipation is bounded below by θ, preventing degenerate behavior in the PDE.

But from Step 4.2.3 the left-hand side is zero, so

$$\theta \int_Q \|\nabla_y u_0\|^2 \, dy \;\leq\; 0. \tag{3.36}$$

Since $\theta > 0$ and $\|\nabla_y u_0\|^2 \geq 0$, we must have [17]:

$$\int_Q \left|\nabla_y u_0\right|^2 dy = 0 \implies \|\nabla_y u_0\|^2 = 0 \quad \text{a.e. in } Q \tag{3.37}$$

which indicates u_0 is constant in y. By periodicity the constant is the same throughout Q, so we write:

$$u_0(x, y) = u_0(x). \tag{3.38}$$

Step 5: Solve the $\mathcal{O}(\varepsilon^{-1})$ Equation

5.1 **Start with the $\mathcal{O}(\varepsilon^{-1})$ equation.** From the asymptotic expansion

$$\frac{1}{\varepsilon^2} L_1 u_0 \;+\; \frac{1}{\varepsilon} \left(L_1 u_1 + L_2 u_0\right) \;+\; \mathcal{O}(1) \;=\; f(x). \tag{3.39}$$

At order $\mathcal{O}(\varepsilon^{-1})$, we obtain [18]:

$$L_1 u_1 + L_2 u_0 = 0. \tag{3.40}$$

5.2 **Simplify $L_2 u_0$.** Since $u_0 = u_0(x)$ is independent of the fast variable y, recall

$$L_2 \;=\; -\sum_{i,j} \left[\frac{\partial}{\partial y_j} \left(a^{ij}(y) \, \frac{\partial}{\partial x_i} \right) \;+\; \frac{\partial}{\partial x_j} \left(a^{ij}(y) \, \frac{\partial}{\partial y_i} \right) \right]. \tag{3.41}$$

When applied to $u_0(x)$:

① The first term:

$$\frac{\partial}{\partial y_j} \left(a^{ij}(y) \, \frac{\partial u_0}{\partial x_i} \right) = \frac{\partial a^{ij}(y)}{\partial y_j} \, \frac{\partial u_0}{\partial x_i}. \tag{3.42}$$

since $\partial u_0 / \partial x_i$ is independent of y.

② The second term:

$$\frac{\partial}{\partial x_j} \left(a^{ij}(y) \, \frac{\partial u_0}{\partial y_i} \right) = 0. \tag{3.43}$$

because $\partial u_0 / \partial y_i = 0$.

[17] Physics insight: The energy $\int_Q |\nabla_y u_0|^2 dy$ represents microscopic oscillations. Uniform ellipticity ($A(y) \geq \theta I$) makes this energy non-negative, so $L_1 u_0 = 0$ kills all oscillations.
Mathematics: This forces u_1 to compensate via the cell problem: $-\nabla_y \cdot (A(y) \nabla_y \chi^k) = \nabla_y \cdot (A(y) e_k)$, where $\chi^k(y)$ encodes microstructure effects.

[18] Same as in the case of $\mathcal{O}(\varepsilon^{-2})$, the term $L_1 u_1 + L_2 u_0$ has to be 0. Otherwise, the system would blow up, as $\varepsilon \to \infty$.

Hence

$$L_2 u_0 = -\sum_{i,j} \frac{\partial a^{ij}(y)}{\partial y_j} \frac{\partial u_0}{\partial x_i}. \tag{3.44}$$

5.3 **Equation for u_1.** Substitute into $L_1 u_1 + L_2 u_0 = 0$. Recall

$$L_1 u_1 = -\sum_{i,j} \frac{\partial}{\partial y_j} \left(a^{ij}(y) \frac{\partial u_1}{\partial y_i} \right). \tag{3.45}$$

Thus

$$-\sum_{i,j} \frac{\partial}{\partial y_j} \left(a^{ij}(y) \frac{\partial u_1}{\partial y_i} \right) - \sum_{i,j} \frac{\partial a^{ij}(y)}{\partial y_j} \frac{\partial u_0}{\partial x_i} = 0. \tag{3.46}$$

Rearranged:

$$-\sum_{i,j} \frac{\partial}{\partial y_j} \left(a^{ij}(y) \frac{\partial u_1}{\partial y_i} \right) = \sum_{i,j} \frac{\partial a^{ij}(y)}{\partial y_j} \frac{\partial u_0}{\partial x_i}. \tag{3.47}$$

5.4 **Ansatz for u_1.** Seek a separated form decoupling microscale and macroscale:

$$u_1(x, y) = -\sum_{k=1}^{n} \chi^k(y) \frac{\partial u_0}{\partial x_k}(x) + \widetilde{u}_1(x). \tag{3.48}$$

where

i. $\chi^k(y)$ are the corrector (cell) functions, depending only on y.

ii. $\widetilde{u}_1(x)$ is a purely macroscopic term (independent of y).

5.5 **Cell problem for $\chi^k(y)$.** Substitute the ansatz into the PDE for u_1:

1. Compute the y-derivative:
$$\frac{\partial u_1}{\partial y_i} = -\sum_{k=1}^{n} \frac{\partial \chi^k(y)}{\partial y_i} \frac{\partial u_0}{\partial x_k}. \tag{3.49}$$

2. Substitute into $-\sum_{i,j} \frac{\partial}{\partial y_j} \left(a^{ij}(y)\, \partial_{y_i} u_1 \right) = \sum_{i,j} \partial_{y_j} a^{ij}(y)\, \partial_{x_i} u_0$. The left becomes
$$-\sum_{i,j} \frac{\partial}{\partial y_j} \left(a^{ij}(y) \left[-\sum_k \partial_{y_i} \chi^k(y)\, \partial_{x_k} u_0 \right] \right) = \sum_{i,j} \frac{\partial}{\partial y_j} \left(a^{ij}(y) \sum_k \frac{\partial \chi^k}{\partial y_i} \right) \frac{\partial u_0}{\partial x_k}. \tag{3.50}$$

3. Swap summations $\sum_{i,j} \sum_k$ and match with the right-hand side $\sum_{i,j} \partial_{y_j} a^{ij}(y)\, \partial_{x_i} u_0$. Since $\partial_{x_k} u_0$ is arbitrary for each k, require for each fixed k:
$$-\sum_{i,j} \frac{\partial}{\partial y_j} \left(a^{ij}(y) \frac{\partial \chi^k(y)}{\partial y_i} \right) = \sum_j \frac{\partial a^{kj}(y)}{\partial y_j}. \tag{3.51}$$

This is the cell problem for $\chi^k(y)$.

5.6 **Normalization and uniqueness for χ^k.** To fix the arbitrary additive constant in χ^k, impose:

i. Periodicity: $\chi^k(y)$ is Q-periodic in y.

ii. Zero mean: $\int_Q \chi^k(y)\, dy = 0$.

5.7 **Physical interpretation.**

i. The correctors $\chi^k(y)$ capture how microscale variations in $a^{ij}(y)$ perturb the leading-order solution relative to the macroscopic gradient $\nabla_x u_0$.

ii. The eventual homogenized (effective) equation at order $\mathcal{O}(1)$ will involve averaged coefficients built from $a^{ij}(y)$ and $\chi^k(y)$.

Step 6: Solve the $\mathcal{O}(1)$ Equation (Homogenized PDE)

6.1 **Collect $\mathcal{O}(1)$ terms.** From the asymptotic expansion one has

$$L_1 u_2 + L_2 u_1 + L_3 u_0 = f(x). \tag{3.52}$$

Here:

i. $L_1 u_2$: microscopic operator acting on u_2.

ii. $L_2 u_1$: coupling between micro/macroscale via u_1.

iii. $L_3 u_0$: macroscopic operator acting on u_0.

6.2 **Average over the unit cell Q.** Integrate the $\mathcal{O}(1)$ equation over Q and use periodicity:

$$\int_Q L_1 u_2\, dy \;+\; \int_Q L_2 u_1\, dy \;+\; \int_Q L_3 u_0\, dy \;=\; \int_Q f(x)\, dy. \tag{3.53}$$

By the periodic divergence theorem (boundary terms cancel) and since L_1 involves $\partial_{y_j}(\cdot)$ on a periodic cell,

$$\int_Q L_1 u_2\, dy = 0. \tag{3.54}$$

Hence the averaged equation reduces to

$$\int_Q L_2 u_1\, dy \;+\; \int_Q L_3 u_0\, dy \;=\; \int_Q f(x)\, dy. \tag{3.55}$$

6.3 **Compute $\int_Q L_3 u_0\, dy$.** Since $u_0 = u_0(x)$ is independent of y, recall the form of L_3:

$$L_3 u_0 \;=\; -\sum_{i,j} \frac{\partial}{\partial x_j}\Big(a^{ij}(y)\, \frac{\partial u_0}{\partial x_i}\Big). \tag{3.56}$$

Integrating over Q:

$$\int_Q L_3 u_0\, dy = -\sum_{i,j} \int_Q \frac{\partial}{\partial x_j}\Big(a^{ij}(y)\, \frac{\partial u_0}{\partial x_i}\Big)\, dy = -\sum_{i,j}\Big(\int_Q a^{ij}(y)\, dy\Big)\, \frac{\partial^2 u_0}{\partial x_i \partial x_j}. \tag{3.57}$$

since ∂_{x_j} can be pulled outside the y-integral, and u_0 depends only on x.

6.4 **Compute $\int_Q L_2 u_1 \, dy$.** Recall

$$L_2 = -\sum_{i,j} \Big[\frac{\partial}{\partial y_j} \Big(a^{ij}(y) \, \frac{\partial}{\partial x_i} \Big) \; + \; \frac{\partial}{\partial x_j} \Big(a^{ij}(y) \, \frac{\partial}{\partial y_i} \Big) \Big]. \tag{3.58}$$

and the ansatz for u_1 from Step 5:

$$u_1(x,y) = -\sum_{k=1}^{n} \chi^k(y) \, \frac{\partial u_0}{\partial x_k}(x) \; + \; \widetilde{u}_1(x). \tag{3.59}$$

where χ^k are cell functions (depending only on y), and $\widetilde{u}_1(x)$ is independent of y. Split $L_2 u_1$ into two terms:

$$L_2 u_1 = -\sum_{i,j} \frac{\partial}{\partial y_j} \Big(a^{ij}(y) \, \frac{\partial u_1}{\partial x_i} \Big) \; - \; \sum_{i,j} \frac{\partial}{\partial x_j} \Big(a^{ij}(y) \, \frac{\partial u_1}{\partial y_i} \Big). \tag{3.60}$$

i. Term 1: $\int_Q \frac{\partial}{\partial y_j} \Big(a^{ij}(y) \, \frac{\partial u_1}{\partial x_i} \Big) \, dy$. Since

$$\frac{\partial u_1}{\partial x_i} = -\sum_k \chi^k(y) \, \frac{\partial^2 u_0}{\partial x_i \partial x_k} \; + \; \frac{\partial \widetilde{u}_1}{\partial x_i}. \tag{3.61}$$

and χ^k depends only on y, $\partial^2 u_0 / \partial x_i \partial x_k$ is independent of y. By periodicity in y.

$$\int_Q \frac{\partial}{\partial y_j} \Big(a^{ij}(y) \, \frac{\partial u_1}{\partial x_i} \Big) \, dy = 0. \tag{3.62}$$

ii. Term 2: $\int_Q \frac{\partial}{\partial x_j} \Big(a^{ij}(y) \, \frac{\partial u_1}{\partial y_i} \Big) \, dy$. First compute

$$\frac{\partial u_1}{\partial y_i} = -\sum_{k=1}^{n} \frac{\partial \chi^k(y)}{\partial y_i} \, \frac{\partial u_0}{\partial x_k}, \tag{3.63}$$

since $\widetilde{u}_1(x)$ is independent of y. Thus

$$\int_Q \frac{\partial}{\partial x_j} \Big(a^{ij}(y) \, \frac{\partial u_1}{\partial y_i} \Big) \, dy = \int_Q \frac{\partial}{\partial x_j} \Big(-\sum_k a^{ij}(y) \, \frac{\partial \chi^k(y)}{\partial y_i} \, \frac{\partial u_0}{\partial x_k} \Big) \, dy. \tag{3.64}$$

Pull ∂_{x_j} outside the y-integral:

$$= -\sum_k \Big(\int_Q a^{ij}(y) \, \frac{\partial \chi^k(y)}{\partial y_i} \, dy \Big) \, \frac{\partial^2 u_0}{\partial x_j \partial x_k}. \tag{3.65}$$

Combining both terms and the overall minus sign in L_2:

$$\int_Q L_2 u_1 \, dy = \sum_{i,j,k} \Big(\int_Q a^{ij}(y) \, \frac{\partial \chi^k(y)}{\partial y_i} \, dy \Big) \, \frac{\partial^2 u_0}{\partial x_j \partial x_k}. \tag{3.66}$$

6.5 **Combine terms and define homogenized coefficients.** Substitute the results of 6.3 and 6.4 into the averaged equation from 6.2:

$$\sum_{i,j,k} \Big(\int_Q a^{ij}(y) \, \frac{\partial \chi^k(y)}{\partial y_i} \, dy \Big) \, \frac{\partial^2 u_0}{\partial x_j \partial x_k} \; - \; \sum_{i,j} \Big(\int_Q a^{ij}(y) \, dy \Big) \, \frac{\partial^2 u_0}{\partial x_i \partial x_j} \; = \; \int_Q f(x) \, dy. \tag{3.67}$$

If $|Q| = 1$ (unit cell volume), then $\int_Q f(x) \, dy = f(x)$. Next, rewrite the first sum by swapping indices $i \leftrightarrow j$:

$$\sum_{j,k} \Big(\int_Q a^{ji}(y) \, \frac{\partial \chi^k(y)}{\partial y_i} \, dy \Big) \, \frac{\partial^2 u_0}{\partial x_j \partial x_k} \; - \; \sum_{i,j} \Big(\int_Q a^{ij}(y) \, dy \Big) \, \frac{\partial^2 u_0}{\partial x_i \partial x_j} \; = \; f(x). \tag{3.68}$$

Combine into a single sum over j, k. Observe that in the second term the dummy indices i, j can be renamed to j, k as well (since it sums over all indices). We form the bracket $\int_Q \big(\sum_i a^{ji}(y) \, \partial_{y_i} \chi^k(y) \; - \; a^{jk}(y) \big) dy$. Define the homogenized coefficients

$$\bar{a}^{\,jk} \; = \; \int_Q \Big(a^{jk}(y) \; - \; \sum_{i=1}^{n} a^{ji}(y) \, \frac{\partial \chi^k(y)}{\partial y_i} \Big) \, dy. \tag{3.69}$$

Then the averaged equation becomes

$$- \sum_{j,k} \bar{a}^{\,jk} \, \frac{\partial^2 u_0}{\partial x_j \partial x_k} \; = \; f(x). \tag{3.70}$$

6.6 **State the homogenized PDE (with boundary condition).** The leading-order term $u_0(x)$ satisfies the homogenized equation

$$- \sum_{j,k=1}^{n} \bar{a}^{\,jk} \, \frac{\partial^2 u_0}{\partial x_j \partial x_k} = f(x), \quad u_0|_{\partial \Omega} = 0 \quad \text{(or other imposed BCs)}. \tag{3.71}$$

Here $\bar{a}^{\,jk}$ are the effective coefficients computed from the cell functions $\chi^k(y)$ as above.

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